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Topological Coalgebras

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INTRODUCTION

The basic theory of coalgebras over fields is developed by Sweedler [2], and as he shows, colalgebras over fields have many pretty properties comparatively with algebras. For instance, any cocommutative coalgebra is the direct sum of its irreducible components, and as the author [3] shows, a comodule is (an) injective (cogenerator) if and only (faithfully) coflat. This structural simplicity of coalgebras is due to the finiteness theorem which tells that any coalgebra is the union of its finite dimensional subcoalgebras. This theorem reduces most of the study of coalgebras to the study of finite dimensional algebras. Using such reduction, Sweedler shows the existence of several (co)universal coalgebras such as the cofree coalgebra or the universal measuring coalgebra. The construction is, however, never easy unlike the construction of free algebras or polynomial algebras. Sometimes we have no counterpart in coalgebra of such basic universal objects in algebra theory as the endomorphism ring or the localization.

With this paper, we begin the study of topological coalgebras. We consider usual coalgebras as a special case of topological coalgebras, namely, discrete coalgebras. To define a topological coalgebra, we have only to consider a topological vector space instead of a vector space, and the complete tensor product $\hat{\otimes}$ instead of the tensor product. The basic part of the theory goes quite similarly to coalgebra theory, but of course the finiteness theorem does not hold, because the category of topological coalgebras contains the opposite category of all algebras. That is, if A is an algebra, the dual space A^* has a topological coalgebra structure with comultiplication $\Delta\colon A^* \to A^* \hat{\otimes} A^*$ $(= (A \otimes A)^*)$ induced from the multiplication. If A and B are two algebras, there is a one-to-one correspondence between algebra maps $A \to B$ and topological coalgebra maps $B^* \to A^*$. Thus the category of topological coalgebras is big enough to contain the category of coalgebras and the opposite category of algebras.

In this paper we present the foundation of the theory of topological

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coalgebras and construct several basic (co)universal objects in topological coalgebra theory. We deal with the colocalization, the universal measuring topological coalgebra, and the coendomorphism topological coalgebra. Not only do such objects exist, but also one sees that they are constructed in a very easy and naive way unlike coalgebra theory.

The first section is devoted to the foundation of topological vector spaces and topological coalgebras. As far as the author's knowledge, there is no reference on topological coalgebras.

In Section 2, we deal with colocalization. Let C be a cocommutative complete topological coalgebra and let S be a multiplicative set of continuous linear forms on C. We show there is a topological coalgebra $C_{[S]}$ together with a continuous coalgebra map $C_{[S]} \to C$ such that every element in S is invertible as a linear form on $C_{[S]}$ and such that it is couniversal relative to this property. In addition, the coalgebra $C_{[S]}$ is constructed as a projective limit, quite dually to the construction of localization in commutative algebra. For the proof, the commutativity of the completre tensor product with projective limits is used. If A is a commutative algebra and S a multiplicative subset of A, then $A^*_{[S]}$ is precisely the dual of the localization $A[S^{-1}]$. Hence we call $C_{[S]}$ the colocalization of C relative to S. In case C is discrete and C^* is a domain, we can take $C^* - \{0\}$ as S. The colocalization $C_{[C^* - \{0\}]}$ is the dual concept to the total ring of fractions.

In Section 3, we deal with the universal measuring topological coalgebra. Recall that if A is an algebra and C a coalgebra, then $\mathrm{Hom}(C, A)$ is an algebra with convolution product. For two algebras A and B, the universal measuring coalgebra $\mathrm{Mes}(A, B)$ is such a coalgebra that there is a one-to-one correspondence between coalgebra maps $C \to \mathrm{Mes}(A, B)$ and algebra maps $A \to \mathrm{Hom}(C, B)$. We have a quite similar situation with topological coalgebras. Let C be a topological coalgebra. Then the vector space of all continuous linear maps $C \to A$ form an algebra $\mathrm{Hom}_c(C, A)$. (We view A as a discrete space). There is also a topological coalgebra $\mathrm{Mes}_c(A, B)$ such that there is a one-to-one correspondence between topological coalgebra maps $C \to \mathrm{Mes}_c(A, B)$ and algebra maps $A \to \mathrm{Hom}_c(C, B)$ for any topological coalgebra C. In addition, the topological coalgebra $\mathrm{Mes}_c(A, B)$ is realized as the space of all homomorphisms of complexes $A^{\cdot} \to B^{\cdot}$ where $A^{\cdot}$ denotes the following complex of vector spaces with arrows induced from the algebra structure on A:

$$k \to A \;\substack{\rightarrow \\ \rightarrow \\ \leftarrow}\; A \otimes A \;\substack{\rightarrow \\ \rightarrow \\ \rightarrow \\ \leftarrow \\ \leftarrow}\; A \otimes A \otimes A \;\substack{\rightarrow \\ \rightarrow \\ \rightarrow \\ \rightarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow}\; \cdots$$

In Section 4, we deal with the coendomorphism topological coalgebra.

We begin with construction of the co-hom topological vector space $c(W, V)$ for topological vector spaces V and W. This satisfies the following universal mapping property: for any topological vector space C, there is a one-to-one correspondence between continuous linear maps $V \to W \hat{\otimes} C$ and continuous linear maps $c(W, V) \to \hat{C}$. In addition, the structure of $c(W, V)$ is very explicit just as in the previous sections, namely, of the form $W^{\odot} \otimes V$ ($W^{\odot}$ being the continuous dual of W) with a special topology called the Ω-topology, hence written as $W^{\odot} \otimes_{\Omega} V$. It follows from the universal mapping property that $c(V, V)$ has a topological coalgebra structure. Its continuous dual is identified with the algebra of all continuous linear endomorphisms of $\hat{V}$.

In Section 5, we study the Ω-topology more closely. It is shown that for a topological coalgebra C, the multiplication on $C^{\odot}$ is a continuous map $C^{\odot} \otimes_{\Omega} C^{\odot} \to C^{\odot}$. Hence we call $C^{\odot}$ the Ω-topological algebra associated with C. We establish some correspondence between topological C-comodules and Ω-topological $C^{\odot}$-modules. We also show that if V is a right topological C-comodule, then $V^{\odot}$ is a left topological C-comodule. We will clarify the relationship between this duality and the previous correspondence between topological comodules and Ω-topological modules.

1. Topological Vector Spaces and Topological Coalgebras

All the statements on topological vector spaces in this section follow from the theory of topological groups, in fact from the very elementary part. We refer to [1] for the theory of topological groups.

We work over a fixed field k, and denote $\otimes_k$ and Hom_k by $\otimes$ and Hom. We use the term "subspace" to mean a linear subspace of a vector space, not a subset of a topological space.

A topology on a vector space is said to be *linear* if every translation $v \mapsto v + w$ is continuous and if there is a basis of neighborhoods of 0 consisting of subspaces. Such a basis of neighborhoods is called a basis of the linear topology. Any directed set of subspaces becomes a basis of a uniquely determined linear topology.

We define a *topological vector space* to be a vector space with a linear topology on it. A basis of the linear topology is called a *topological basis* of the topological vector space, while a free basis is called a *linear basis*. We give k the discrete topology.

A linear map of topological vector spaces is continuous if the inverse image of every member of a topological basis of the range is an open subspace of the domain.

The closure of a subset A of a topological vector space is denoted by $\bar{A}$

and it is given as the intersection $\bigcap_\alpha (A + V_\alpha)$ if the topological vector space has a topological basis $\{V_\alpha\}_\alpha$. In particular the space is separated if and only if $\bigcap_\alpha V_\alpha = 0$, or $\{0\}$ is closed.

There are several basic formations to have a topological vector space from given topological vector spaces.

1.1. *Subspaces and Quotient Spaces*

Let V be a topological vector space with a topological basis $\{V_\alpha\}_\alpha$. Let W be a subspace of V. Then W is a topological vector space with the induced topology, and so is V/W with the quotient topology. Thus W and V/W have topological bases $\{W \cap V_\alpha\}_\alpha$ and $\{(W + V_\alpha)/W\}_\alpha$, respectively.

1.2. *Direct Products and Sums*

Assume we are given a family of topological vector spaces $V^{(p)}$, $p \in P$. The direct product $\prod_p V^{(p)}$ is a topological vector space with the direct product topology. Thus if $\mathscr{B}^{(p)}$ is a topological basis of $V^{(p)}$ for each p, then the products $\prod_p U^{(p)}$ with $U^{(p)} \in \mathscr{B}^{(p)}$, and $U^{(p)} = V^{(p)}$ except a finite number of p form a topological basis for $\prod_p V^{(p)}$. If we are given a subspace $W^{(p)}$ of $V^{(p)}$ for each p, then the direct product topology on $\prod_p W^{(p)}$ is induced from $\prod_p V^{(p)}$. On the other hand, we define the direct sum topology on $\bigoplus_p V^{(p)}$ by taking all direct sums $\bigoplus_p U^{(p)}$ with $U^{(p)} \in \mathscr{B}^{(p)}$ as a topological basis. Of course, this topology is not induced from the direct product.

1.3. Hom(M, V)

Let M be a vector space and let V be a topological vector space. Then Hom(M, V) is the direct product of dim M (possibly infinite) copies of V. Hence it has the direct product topology. In particular, M^* $(= \mathrm{Hom}(M, k))$ is a topological vector space.

1.4. *Projective and Inductive Limits*

We mean by a projective (resp. an inductive) system in a category **A** a contravariant (resp. covariant) functor from some small category into **A**. Such a system is simply denoted as $\{V^{(p)}\}_p$ with the index category omitted. So, this symbol means a contravariant (resp. covariant) functor $p \mapsto V^{(p)}$.

Let $\{V^{(p)}\}_p$ be a projective system of topological vector spaces. Recall the definition of the projective limit $\varprojlim_p V^{(p)}$. It is the subspace of all elements $(v_p)_p$ in $\prod_p V^{(p)}$ such that for any map $f\colon q \to r$ in the index category, the element v_r is mapped to v_q by the induced continuous linear map $f\colon V^{(r)} \to V^{(q)}$. We give it the induced topology from the direct product $\prod_p V^{(p)}$, and call it the projective limit topological vector space. If every

$V^{(p)}$ is separated, then it is a closed subspace of the direct product. The inductive limit $\varinjlim_i W^{(i)}$ of an inductive system $\{W^{(i)}\}_i$ of topological vector spaces is defined dually. It is given the quotient topology of the direct sum $\bigoplus_i W^{(i)}$, and called the inductive limit topological vector space.

1.5. *The Tensor Product*

Let $V^{(1)}$ and $V^{(2)}$ be two topological vector spaces with topological bases $\{V_\alpha^{(1)}\}_\alpha$ and $\{V_\gamma^{(2)}\}_\gamma$ respectively. There is a linear topology called the tensor product topology on $V^{(1)} \otimes V^{(2)}$ having the set

$$\{V_\alpha^{(1)} \otimes V^{(2)} + V^{(1)} \otimes V_\gamma^{(2)}\}_{\alpha,\gamma}$$

as a basis. If $W^{(i)}$ is a subspace of $V^{(i)}$ for $i = 1, 2$, the tensor product topology on $W^{(1)} \otimes W^{(2)}$ is induced from $V^{(1)} \otimes V^{(2)}$. More generally, for a finite number of topological vector spaces $V^{(1)},..., V^{(n)}$, the tensor product topology on $V^{(1)} \otimes \cdots \otimes V^{(n)}$ is defined in the same way. We have the associativity $(V^{(1)} \otimes V^{(2)}) \otimes V^{(3)} \cong V^{(1)} \otimes V^{(2)} \otimes V^{(3)} \simeq V^{(1)} \otimes (V^{(2)} \otimes V^{(3)})$ and the commutativity $V^{(1)} \otimes V^{(2)} \simeq V^{(2)} \otimes V^{(1)}$ as topological vector spaces.

Let V be a topological vector space with a topological basis $\{V_\alpha\}_\alpha$. The projective limit $\varprojlim_\alpha V/V_\alpha$, where every V/V_α is discrete, is called the *completion* of V and denoted by $\hat{V}$. It has a topological basis $\{(\hat{V})_\alpha\}_\alpha$, where $(\hat{V})_\alpha$ denotes the kernel of the projection $\hat{V} \to V/V_\alpha$. The projections $V \to V/V_\alpha$ induce a continuous linear map $V \to \hat{V}$ which induces bijections $V/V_\alpha \simeq \hat{V}/(\hat{V})_\alpha$. Hence V has a dense image in $\hat{V}$ and every open subspace of V is the inverse image of some open subspace of $\hat{V}$. V is separated if and only if the map $V \to \hat{V}$ is injective. V is called *complete* if $V \simeq \hat{V}$ under the canonical map. It follows that $\hat{V}$ is complete.

Any continuous linear map f of topological vector spaces induces a continuous linear map of the completions which we denote by $\hat{f}$. If W is a subspace (with the induced topology) of the space V, the inclusion $W \to V$ induces an isomorphism of $\hat{W}$ onto the kernel of the map $\hat{V} \to \widehat{V/W}$ which is a closed subspace of $\hat{V}$. Since W has a dense image in $\hat{W}$, it follows that $\hat{W}$ is the closure of the image of W in $\hat{V}$. Hence closed subspaces of a complete vector space are complete. W is dense in V if and only if $\hat{W} = \hat{V}$, or $\widehat{V/W} = 0$.

We show the completion commutes with the direct product:

1.6. Proposition. *Let $V^{(p)}$, $p \in P$, be a set of topological vector spaces. We have*

$$\widehat{\prod_p V^{(p)}} = \prod_p \widehat{V^{(p)}}.$$

Proof. Let $\{V_x^{(p)} \mid x \in A(p)\}$ be a topological basis of $V^{(p)}$. We have by definition

$$\widehat{\prod_p V^{(p)}} = \varprojlim_{(F,\alpha)} \prod_{p\in F} (V^{(p)}/V^{(p)}_{\alpha(p)}),$$

where $\varprojlim$ is taken over all pairs (F, α) of a finite subset F of P and $\alpha = \{\alpha(p) \mid p \in F\}$ an element in $\prod_{p\in F} A(p)$. We can factor this $\varprojlim$ as

$$\varprojlim_{(F,\alpha)} = \varprojlim_F \circ \varprojlim_\alpha ,$$

where $\varprojlim_\alpha$ is taken over all α in the set $\prod_{p\in F} A(p)$. Hence we have

$$\widehat{\prod_p V^{(p)}} = \varprojlim_F \varprojlim_\alpha \prod_{p\in F} (V^{(p)}/V^{(p)}_{\alpha(p)}) = \varprojlim_F \prod_{p\in F} \widehat{V^{(p)}} = \prod_{p\in P} \widehat{V^{(p)}}. \qquad \text{Q.E.D.}$$

1.7. Corollary. *Direct products of complete vector spaces are complete. Projective limits of complete vector spaces are complete.*

1.8. Corollary. *For a vector space M and a topological vector space V, the completion of* $\mathrm{Hom}(M, V)$ *is* $\mathrm{Hom}(M, \hat{V})$.

The completion of the tensor product $V^{(1)} \otimes V^{(2)}$ is denoted by $V^{(1)} \hat{\otimes} V^{(2)}$. More generally, $V^{(1)} \hat{\otimes} \cdots \hat{\otimes} V^{(n)}$ denoted the completion of $V^{(1)} \otimes \cdots \otimes V^{(n)}$. The following identities follow easily from definition:

$$V^{(1)} \hat{\otimes} V^{(2)} = \widehat{V^{(1)}} \hat{\otimes} \widehat{V^{(2)}},$$

$$V \hat{\otimes} k = \hat{V} = k \hat{\otimes} V, \tag{1.9}$$

$$V^{(1)} \hat{\otimes} V^{(2)} \hat{\otimes} V^{(3)} = V^{(1)} \hat{\otimes} (V^{(2)} \hat{\otimes} V^{(3)}) = V^{(1)} \hat{\otimes} (V^{(2)} \otimes V^{(3)}) \quad \text{(etc.)}.$$

We now proceed into the definition of topological coalgebras. A *topological coalgebra* is a triple (C, Δ, ε) consisting of a topological vector space C and continuous linear maps $\Delta: C \to C \hat{\otimes} C$, $\varepsilon: C \to k$ satisfying:

1.10. Coassociativity. The following composites are the same.

$$C \xrightarrow{\Delta} C \hat{\otimes} C \xrightarrow{\Delta \hat{\otimes} I} C \hat{\otimes} C \hat{\otimes} C,$$

$$C \xrightarrow{\Delta} C \hat{\otimes} C \xrightarrow{I \hat{\otimes} \Delta} C \hat{\otimes} C \hat{\otimes} C.$$

1.11. Counit Condition. The following composites are the canonical map $C \to \hat{C}$.

$$C \xrightarrow{\Delta} C \hat{\otimes} C \xrightarrow{I \hat{\otimes} \varepsilon} C \hat{\otimes} k = \hat{C},$$

$$C \xrightarrow{\Delta} C \hat{\otimes} C \xrightarrow{\varepsilon \hat{\otimes} I} k \hat{\otimes} C = \hat{C}.$$

Maps of topological coalgebras are continuous linear maps commuting with the structures Δ, ε. The completion $\hat{C}$ of a topological coalgebra C is a topological coalgebra with structures $\hat{\Delta}$ and $\hat{\varepsilon}$, and the canonical map $C \to \hat{C}$ is a continuous coalgebra map. If V is a subspace of C, we have $V \hat{\otimes} V \subset C \hat{\otimes} C$. Hence if $\Delta(V) \subset V \hat{\otimes} V$, V has the structure of a topological coalgebra. Such coalgebras are called subcoalgebras. For instance, C is a subcoalgebra of $\hat{C}$ if C is separated.

Discrete topological coalgebras are precisely coalgebras in the usual sense. As another basic example, we will see the dual A^* of an algebra A has the structure of a topological coalgebra. To see this we need the fact that the completed tensor product $\hat{\otimes}$ commutes with the direct product as specified in the following:

1.12. THEOREM. *Let $V^{(p)}$, $p \in P$, and W be topological vector spaces. Collecting all continuous linear maps*

$$\mathrm{proj}_q \hat{\otimes} I \colon \left(\prod_p V^{(p)}\right) \hat{\otimes} W \to V^{(q)} \hat{\otimes} W$$

we get an isomorphism of topological vector spaces:

$$\left(\prod_p V^{(p)}\right) \hat{\otimes} W \simeq \prod_p (V^{(p)} \hat{\otimes} W).$$

Proof. Before completion, collecting maps

$$\mathrm{proj}_q \otimes I \colon \left(\prod_p V^{(p)}\right) \otimes W \to V^{(q)} \otimes W$$

gives a continuous linear map

$$\theta \colon \left(\prod_p V^{(p)}\right) \otimes W \to \prod_p (V^{(p)} \otimes W).$$

This map is injective as easily seen and induces the map in the theorem after completion (Use 1.6). To specify the completion, let $\{V_x^{(p)} \mid x \in A(p)\}$ be a topological basis of $V^{(p)}$ and $\{W_\gamma \mid \gamma \in \Gamma\}$ a topological basis of W. For a finite subset F of P, an element $\alpha = \{\alpha(p) \mid p \in F\}$ in $\prod_{p \in F} A(p)$, and $\gamma \in \Gamma$, we have a canonical isomorphism

$$\left(\prod_{p \in F} (V^{(p)}/V^{(p)}_{\alpha(p)})\right) \otimes (W/W_\gamma) \simeq \prod_{p \in F} ((V^{(p)}/V^{(p)}_{\alpha(p)}) \otimes (W/W_\gamma)).$$

It follows from the definition that the completion $\hat{\theta}$ is obtained by taking $\varprojlim$ of this isomorphism over all such triples (F, α, γ). This proves the claim. Q.E.D.

1.13. COROLLARY. *Let $V^{(p)}$, $p \in P$, and $W^{(q)}$, $q \in Q$, be families of topological vector spaces. The maps*

$$\mathrm{proj}_p \hat{\otimes} \mathrm{proj}_q : \left(\prod_p V^{(p)}\right) \hat{\otimes} \left(\prod_q W^{(q)}\right) \to V^{(p)} \hat{\otimes} W^{(q)}$$

induce an isomorphism of topological vector spaces

$$\left(\prod_p V^{(p)}\right) \hat{\otimes} \left(\prod_q W^{(q)}\right) \simeq \prod_{p,q} (V^{(q)} \hat{\otimes} W^{(q)}).$$

1.14. COROLLARY. *Let M, N be vector spaces and V, W be topological vector spaces. There is a canonical isomorphism of topological vector spaces*

$$\mathrm{Hom}(M, V) \hat{\otimes} \mathrm{Hom}(N, W) \simeq \mathrm{Hom}(M \otimes N, V \hat{\otimes} W).$$

In particular we have $M^* \hat{\otimes} N^* \simeq (M \otimes N)^*$. (See 1.3 for topology on $(\ )^*$.)

1.15. EXAMPLE OF TOPOLOGICAL COALGEBRAS. Let A be an algebra. The dual space A^* is a topological coalgebra with $\Delta: A^* \to A^* \hat{\otimes} A^* \simeq (A \otimes A)^*$ the dual of the multiplication and $\varepsilon: A^* \to k$ the dual of the unit. Any algebra map $A \to B$ induces a continuous coalgebra map $B^* \to A^*$.

Let C be a topological coalgebra. A *topological* (*right*) *comodule* for C is a pair (V, ρ) consisting of a topological vector space V and a continuous linear map $\rho: V \to V \hat{\otimes} C$ satisfying (1.16) and (1.17) below.

1.16. The composites are the same.

$$V \xrightarrow{\rho} V \hat{\otimes} C \xrightarrow{I \hat{\otimes} \Delta} V \hat{\otimes} C \hat{\otimes} C,$$

$$V \xrightarrow{\rho} V \hat{\otimes} C \xrightarrow{\rho \hat{\otimes} I} V \hat{\otimes} C \hat{\otimes} C.$$

1.17. The canonical map $V \to \hat{V}$ is the composite:

$$V \xrightarrow{\rho} V \hat{\otimes} C \xrightarrow{I \hat{\otimes} \varepsilon} V \hat{\otimes} k = \hat{V}.$$

Maps of topological comodules and subcomodules of a topological comodule are defined in an obvious way. If V is a topological comodule for C, the completion $\hat{V}$ is also a topological comodule for C with structure $\hat{\rho}$, and the canonical map $V \to \hat{V}$ is a homomorphism. Since $V \hat{\otimes} C = V \hat{\otimes} \hat{C}$, one sees that topological C-comodules and topological $\hat{C}$-comodules are the same concepts.

If A is an algebra and M a right A-module, it is easy to see M^* is a topological A^*-comodule with structure $M^* \to M^* \hat{\otimes} A^* \simeq (M \otimes A)^*$ the dual of the module structure. In the following, we describe all complete topological A^*-comodules. Let V be a complete topological vector space. We have by (1.14)

$$V \hat{\otimes} A^* \simeq \mathrm{Hom}(A, V).$$

Let $\rho\colon V \to V \hat{\otimes} A^*$ be a continuous linear map, and for each $a \in A$ let $\rho_a\colon V \to V$ be the composite

$$V \xrightarrow{\rho} V \hat{\otimes} A^* \simeq \mathrm{Hom}(A, V) \xrightarrow{\text{evaluation at } a} V.$$

We have a one-to-one correspondence between a continuous linear map ρ and a family of continuous linear maps $\{\rho_a\}_a$ where ρ_a is linear with respect to a.

1.18. PROPOSITION. *ρ satisfies condition* (1.16) *for $C = A^*$ if and only if $\rho_a \circ \rho_b = \rho_{ab}$ for all a, b in A, and satisfies condition* (1.17) *if and only if $\rho_1 = I$ (the identity).*

Proof. If we view ρ as a map $V \to \mathrm{Hom}(A, V)$, the composites in (1.16) become

$$\phi\colon\quad V \xrightarrow{\rho} \mathrm{Hom}(A, V) \xrightarrow{\mathrm{Hom}(m, I)} \mathrm{Hom}(A \otimes A, V),$$

$$\psi\colon\quad V \xrightarrow{\rho} \mathrm{Hom}(A, V) \xrightarrow{\mathrm{Hom}(I, \rho)} \mathrm{Hom}(A, \mathrm{Hom}(A, V)),$$

where m denotes the multiplication of A. Thus we have

$$\phi(v)(a \otimes b) = \rho_{ab}(v), \quad \psi(v)(b)(a) = \rho_a(\rho_b(v))$$

for $v \in V$, $a, b \in A$. The first statement follows from this by the "coherence":

$$\begin{array}{ccc} V \hat{\otimes} (A_1 \otimes A_2)^* \simeq V \hat{\otimes} A_1^* \hat{\otimes} A_2^* \simeq & & \mathrm{Hom}(A_2, V \hat{\otimes} A_1^*) \\ \wr\| & & \wr\| \\ \mathrm{Hom}(A_1 \otimes A_2, V) & \simeq & \mathrm{Hom}(A_2, \mathrm{Hom}(A_1, V)) \end{array}$$

where we put numbers 1, 2 to distuinguish two A's. The second statement is clear since the composite in (1.17) is just ρ_1. Q.E.D.

By a *topological left A-module*, we mean a left A-module and topological vector space such that every element of A acts on it as a continuous linear endomorphism. As a corollary we have:

1.19. THEOREM. *Let A be an algebra. For any complete topological vector space V, there is a one-to-one correspondence between the structures of topological left A-modules on V and the structures of topological right A^*-comodules on it.*

In this way, complete topological right A^*-comodules are identified with complete topological left A-modules. In particular, this is true for discrete comodules and modules.

2. CONTINUOUS DUAL AND COLOCALIZATION

To construct the colocalization of topological coalgebras, we need the commutativity of $\hat{\otimes}$ with projective limits. It is clear that $\hat{\otimes}$ does not preserve general projective limits. For instance, if M is a discrete vector space and M' the space M with the trivial topology (i.e., such that M is the only open subspace), then the kernel of $\hat{M} \to \hat{M}' = \{0\}$ is not the completion of the kernel of the identity $M \to M'$. Thus the functor $k\,\hat{\otimes}$-does not preserve projective limits. We will, however, show that the complete tensor product $\hat{\otimes}$ preserves projective limits of *complete* topological vector spaces. This property will be used to construct the colocalization. In the following complete topological vector spaces are called complete vector spaces.

We already know that $\hat{\otimes}$ commutes with direct products (1.12, 13). Hence it is enough to check whether it commutes with kernels.

2.1. LEMMA. *Let M be a discrete vector space. The functor $-\hat{\otimes} M$ preserves the kernel of a continuous linear map of complete vector spaces.*

Proof. Let X be a linear basis of M. For any topological vector space V, $V \otimes M$ is identified with the direct sum $\bigoplus_X V$ as a vector space (not topologically). If $\{V_\alpha\}_\alpha$ is a topological basis of V, we have

$$V \hat{\otimes} M = \varprojlim_\alpha (V/V_\alpha) \otimes M \simeq \varprojlim_\alpha \bigoplus_X V/V_\alpha.$$

The right-most term is the subspace of elements in $\prod_X \hat{V}$ which are mapped into $\bigoplus_X V/V_\alpha$ by the projection $\prod_X \hat{V} \to V/V_\alpha$ for all α. It follows that if W is a subspace of V, we have

$$W \hat{\otimes} M = (V \hat{\otimes} M) \cap \Big(\prod_X \hat{W}\Big).$$

Let $f\colon V \to V'$ be a continuous linear map of complete vector spaces and let $W = \mathrm{Ker}(f)$. Then $\prod_X W$ is the kernel of the induced map $\prod_X f\colon \prod_X V \to \prod_X V'$. It follows from the above that $W \hat{\otimes} M$ is the kernel of $f \hat{\otimes} I\colon V \hat{\otimes} M \to V' \hat{\otimes} M$. (We know the topology on $W \hat{\otimes} M$ is induced from $V \hat{\otimes} M$.)

Q.E.D.

2.2. THEOREM. *For any topological vector space T, the functor $- \hat{\otimes} T$ preserves the kernel of a continuous linear map of complete vector spaces.*

Proof. If $\{T_\lambda\}_\lambda$ is a topological basis of T, we have

$$V \hat{\otimes} T = \varprojlim_{\lambda} V \hat{\otimes} (T/T_\lambda)$$

for any topological vector space V by definition (T/T_λ being discrete). The claim follows from (2.1) above by the commutativity of projective limits.

Q.E.D.

Since $- \hat{\otimes} T$ commutes with direct products (1.12), we have as a corollary:

2.3. THEOREM. *Let $\{V^{(p)}\}_p$ be a projective system of complete vector spaces and let T be a topological vector space. We have a canonical isomorphism of topological vector spaces*

$$(\varprojlim_{p} V^{(p)}) \hat{\otimes} T \simeq \varprojlim_{p} (V^{(p)} \hat{\otimes} T).$$

Applying the isomorphism twice we get:

2.4. THEOREM. *Let $\{V^{(p)}\}_p$ and $\{W^{(q)}\}_q$ be two projective systems of complete vector spaces. There is a canonical isomorphism of topological vector spaces*

$$(\varprojlim_{p} V^{(p)}) \hat{\otimes} (\varprojlim_{q} W^{(q)}) \simeq \varprojlim_{p,q} (V^{(p)} \hat{\otimes} W^{(q)}).$$

Before proceeding into the colocalization, we need introduce the *continuous dual* of a topological vector space. Let V be a topological vector space. The vector space of all continuous linear maps $V \to k$ is denoted by $V^{\odot}$ and called the continuous dual. Later we give it a topology, but for the time being it is a discrete vector space. If W is another topological vector space, there is an injective linear map $V^{\odot} \otimes W^{\odot} \to (V \hat{\otimes} W)^{\odot}$ sending $f \otimes g$ to $f \hat{\otimes} g\colon V \hat{\otimes} W \to k \hat{\otimes} k = k$. The injectivity follows since $(V \hat{\otimes} W)^{\odot} = (V \otimes W)^{\odot}$. It is worth noting that $V^{\odot}$ is the directed union of $(V/V_\alpha)^*$ for all α if $\{V_\alpha\}_\alpha$ is a topological basis.

If C is a topological coalgebra, the continuous dual $C^{\odot}$ is an algebra with multiplication

$$C^{\odot} \otimes C^{\odot} \xrightarrow{\text{cano}} (C \hat{\otimes} C)^{\odot} \xrightarrow{\Delta^{\odot}} C^{\odot}$$

and unit $\varepsilon^{\odot}: k \simeq k^{\odot} \to C^{\odot}$. If V is a right topological comodule for C, the completion $\hat{V}$ becomes a left (topological) $C^{\odot}$-module, where the multiplication by f in $C^{\odot}$ is defined by the composite:

$$f\cdot: \quad \hat{V} \xrightarrow{\hat{\rho}} V \hat{\otimes} C \xrightarrow{I \hat{\otimes} f} V \hat{\otimes} k = \hat{V}.$$

Since C is a topological comodule for C ($\rho = \Delta$), $\hat{C}$ is a left $C^{\odot}$-module.

Let C be a *cocommutative complete* coalgebra and let S be a multiplicative subset of $C^{\odot}$ containing 1. We associate with S the following small category **S**: The set of objects is S. For p, q in S, the set of morphisms $\mathbf{S}(p, q) = \{r \text{ in } S \mid q = pr\}$. The composition is given by the multiplication in S. We associate with (C, S) a projective system of complete vector spaces $\{C^{(p)}\}_{p \in S}$ as follows: For each p in S, $C^{(p)} = C$. If $r: p \to q$ in **S** (i.e., $q = pr$ in S), the associated map $C^{(q)} \to C^{(p)}$ is the multiplication by r in the $C^{\odot}$-module C. This system has the structure $\{\Delta_{p,q}, \varepsilon_p\}$, where $\Delta_{p,q}: C^{(pq)} \to C^{(p)} \hat{\otimes} C^{(q)}$ is Δ of C, and $\varepsilon_p: C^{(p)} \to k$ is p for any p, q in S. In particular ε_1 is ε of C.

2.5. Lemma. *The system* $\{C^{(p)}, \Delta_{p,q}, \varepsilon_p\}$ *associated with* (C, S) *satisfies conditions*:

(a) *For* $r_i: p_i \to q_i$ $(i = 1, 2)$ *in* **S**, *the diagram commutes*:

$$\begin{array}{ccc} C^{(p_1 p_2)} & \xrightarrow{\Delta_{p_1,p_2}} & C^{(p_1)} \hat{\otimes} C^{(p_2)} \\ \uparrow {\scriptstyle r_1 r_2 \cdot} & & \uparrow {\scriptstyle r_1 \cdot \hat{\otimes} r_2 \cdot} \\ C^{(q_1 q_2)} & \xrightarrow{\Delta_{q_1,q_2}} & C^{(q_1)} \hat{\otimes} C^{(q_2)} \end{array}$$

(b) *For* $r: p \to q$ *in* **S**,

$$\varepsilon_q: C^{(q)} \xrightarrow{r\cdot} C^{(p)} \xrightarrow{\varepsilon_p} k.$$

(c) *For* p, q, r *in* S, *the diagram commutes*:

$$\begin{array}{ccc} C^{(pqr)} & \xrightarrow{\Delta_{pq,r}} & C^{(pq)} \hat{\otimes} C^{(r)} \\ \downarrow {\scriptstyle \Delta_{p,qr}} & & \downarrow {\scriptstyle \Delta_{p,q} \hat{\otimes} I} \\ C^{(p)} \hat{\otimes} C^{(qr)} & \xrightarrow{I \hat{\otimes} \Delta_{q,r}} & C^{(p)} \hat{\otimes} C^{(q)} \hat{\otimes} C^{(r)} \end{array}$$

(d) *For p, q in S,*

$$q\cdot\colon C^{(pq)} \xrightarrow{\Delta_{p,q}} C^{(p)} \hat{\otimes} C^{(q)} \xrightarrow{I \hat{\otimes} \varepsilon_q} C^{(p)}$$

$$p\cdot\colon C^{(pq)} \xrightarrow{\Delta_{p,q}} C^{(p)} \hat{\otimes} C^{(q)} \xrightarrow{\varepsilon_p \hat{\otimes} I} C^{(q)}.$$

Proof. (a) In general, if V is a complete C-comodule, the structure $\rho\colon V \to V \hat{\otimes} C$ is a C-comodule map, hence $(I \hat{\otimes} f\cdot) \circ \rho = \rho \circ (f\cdot)$ for f in $C^{\odot}$. Applying this to $V = C$, we have $(I \hat{\otimes} q\cdot) \circ \Delta = \Delta \circ (q\cdot)$ for q in S, and $(p\cdot \hat{\otimes} I) \circ \Delta = \Delta \circ (p\cdot)$ for p in S by the cocommutativity. It follows that $(p\cdot \hat{\otimes} q\cdot) \circ \Delta = \Delta \circ (pq\cdot)$. (b) For p, q in S, we have

$$pq\colon C \xrightarrow{\Delta} C \hat{\otimes} C \xrightarrow{I \hat{\otimes} q} C \hat{\otimes} k \xrightarrow{p \hat{\otimes} I} k \hat{\otimes} k = k,$$

where $(I \hat{\otimes} q) \circ \Delta$ is $q\cdot$. Hence $p \circ (q\cdot) = pq$. (c) and (d) are clear. Q.E.D.

2.6. THEOREM. *Let C be a cocommutative complete coalgebra and let S be a multiplicative subset of $C^{\odot}$. Let $\{C^{(p)}, \Delta_{p,q}, \varepsilon_p\}_{p,q \in S}$ be the associated projective system. Then*

$$C_{[S]} = \varprojlim_{p} C^{(p)}$$

has the structure of a topological coalgebra with $\Delta_S = \varprojlim_{p,q} \Delta_{p,q}$ and $\varepsilon_S = \varprojlim_p \varepsilon_p$.

Proof. Since $\hat{\otimes}$ commutes with projective limits of complete vector spaces (2.4), we can identify

$$C_{[S]} \hat{\otimes} C_{[S]} = \varprojlim_{p,q} C^{(p)} \hat{\otimes} C^{(q)}.$$

There is a unique continuous linear map $\Delta_S\colon C_{[S]} \to C_{[S]} \hat{\otimes} C_{[S]}$ such that $\mathrm{proj}_{(p,q)} \circ \Delta_S = \Delta_{p,q} \circ \mathrm{proj}_{pq}$ for all p, q in S, by (a) of (2.5). We may write $\Delta_S = \varprojlim_{p,q} \Delta_{p,q}$ by abuse of notation. This Δ_S is coassociative by (c). Put

$$\varepsilon_S\colon C_{[S]} \xrightarrow{\mathrm{proj}_1} C^{(1)} \xrightarrow{\varepsilon_1} k.$$

Then the counit condition holds by (d), and we may write $\varepsilon_S = \varprojlim_p \varepsilon_p$ by (b). Q.E.D.

The proof shows that we have a topological coalgebra as a projective limit from any system $\{C^{(p)}, \Delta_{p,q}, \varepsilon_q\}$ satisfying conditions (a)–(d) of (2.5).

The topological coalgebra $C_{[S]}$ is called the *colocalization* of C with respect to S. There is a canonical continuous coalgebra map

$$\pi\colon C_{[S]} \xrightarrow{\mathrm{proj}_1} C^{(1)} = C.$$

We will clarify the universal mapping property of $(C_{[S]}, \pi)$:

2.7. THEOREM. *Let C be a cocommutative complete coalgebra and let S be a multiplicative subset of $C^{\odot}$.*

(1) *For every p in S, the composite $p \circ \pi\colon C_{[S]} \to k$ is invertible in $(C_{[S]})^{\odot}$.*

(2) *Let $\phi\colon D \to C$ be a continuous homomorphism of topological coalgebras, where D is assumed cocommutative complete. There is a unique continuous homomorphism of topological coalgebras $\bar{\phi}\colon D \to C_{[S]}$ such that $\phi = \pi \circ \bar{\phi}$ if the composite $p \circ \phi\colon D \to k$ is invertible in $D^{\odot}$ for all p in S.*

Proof. We denote by (Δ, ε) (resp. $(\Delta_S, \varepsilon_S)$, resp. $(\Delta_D, \varepsilon_D)$) the coalgebra structure of C (resp. $C_{[S]}$, resp. D).

(1) For p in S, let

$$\bar{p}\colon \quad C_{[S]} \xrightarrow{\pi} C \xrightarrow{p} k,$$

$$x\colon \quad C_{[S]} \xrightarrow{\mathrm{proj}_p} C \xrightarrow{\varepsilon} k.$$

Then we have

$$\begin{aligned}\bar{p}x &= (\bar{p} \hat{\otimes} x) \circ \Delta_S = (p \hat{\otimes} \varepsilon) \circ (\pi \hat{\otimes} \mathrm{proj}_p) \circ \Delta_S = (p \hat{\otimes} \varepsilon) \circ \Delta \circ \mathrm{proj}_p \\ &= p \circ \mathrm{proj}_p = \varepsilon_S\end{aligned}$$

by (a) and (b) of (2.5). Hence $\bar{p}$ is invertible in $(C_{[S]})^{\odot}$ and we have

$$\begin{aligned}\pi \circ (x\cdot) = \pi \circ (I \hat{\otimes} x) \circ \Delta_S &= (I \hat{\otimes} \varepsilon) \circ (\pi \hat{\otimes} \mathrm{proj}_p) \circ \Delta_S \\ &= (I \hat{\otimes} \varepsilon) \circ \Delta \circ \mathrm{proj}_p = I \circ \mathrm{proj}_p = \mathrm{proj}_p.\end{aligned}$$

Hence $\mathrm{proj}_p = \pi \circ (\bar{p}\cdot)^{-1}$.

(2) D is a $D^{\odot}$-module, hence a $C^{\odot}$-module through the algebra map $\phi^{\odot}$. Similarly, $C_{[S]}$ is a $C^{\odot}$-module. If $\bar{\phi}\colon D \to C_{[S]}$ is a continuous coalgebra map such that $\pi \circ \bar{\phi} = \phi$, it is obviously a $C^{\odot}$-module map. Hence for any p in S, we have

$$\mathrm{proj}_p \circ \bar{\phi} = \pi \circ (p\cdot)^{-1} \circ \bar{\phi} = \pi \circ \bar{\phi} \circ (p\cdot)^{-1} = \phi \circ (p\cdot)^{-1}.$$

This means the uniqueness of $\bar{\phi}$. To show the existence, note that the map

$$D \xrightarrow{(p\cdot)^{-1}} D \xrightarrow{\phi} C^{(p)}$$

is natural with respect to p in S. Hence there is a unique continuous linear map $\bar{\phi}\colon D \to C_{[S]}$ such that $\mathrm{proj}_p \circ \bar{\phi} = \phi \circ (p\cdot)^{-1}$ for all p in S. In particular we put $p = 1$ to see $\pi \circ \bar{\phi} = \phi$. We claim that $\bar{\phi}$ is a topological coalgebra map. To prove $\Delta_S \circ \bar{\phi} = (\bar{\phi} \mathbin{\hat{\otimes}} \bar{\phi}) \circ \Delta_D$, let p, q in S. We have

$$\begin{aligned}(\mathrm{proj}_p \mathbin{\hat{\otimes}} \mathrm{proj}_q) \circ \Delta_S \circ \bar{\phi} &= \Delta \circ \mathrm{proj}_{pq} \circ \bar{\phi} = \Delta \circ \phi \circ (pq\cdot)^{-1} \\ &= (\phi \mathbin{\hat{\otimes}} \phi) \circ \Delta_D \circ (pq\cdot)^{-1} = (\phi \mathbin{\hat{\otimes}} \phi) \circ ((p\cdot)^{-1} \mathbin{\hat{\otimes}} (q\cdot)^{-1}) \circ \Delta_D \\ &= (\mathrm{proj}_p \mathbin{\hat{\otimes}} \mathrm{proj}_q) \circ (\bar{\phi} \mathbin{\hat{\otimes}} \bar{\phi}) \circ \Delta_D.\end{aligned}$$

This proves the claim. The identity $\varepsilon_D = \varepsilon_S \circ \bar{\phi}$ follows from $\phi = \pi \circ \bar{\phi}$.

Q.E.D.

For example, let A be a commutative algebra and S a multiplicative subset of A. Obviously A is viewed as a subalgebra of $(A^*)^{\odot}$. (Later we will see they coincide). Since our construction of the colocalization is completely dual to the localization, and the dual of inductive limits gives projective limits, it follows that the colocalization $A^*_{[S]}$ is precisely the dual of the localization $A[S^{-1}]$. Thus we have

$$A^*_{[S]} = (A[S^{-1}])^*.$$

3. The Cofree and the Universal Measuring Topological Coalgebras

An advantage of the topological approach to coalgebras is that we have an easier construction of some universal (or couniversal) objects. We begin with an easy construction of the cofree topological coalgebra on a topological vector space. Let V be a topological vector space. For any integer $n > 1$, let

$$V^{[n]} = V \mathbin{\hat{\otimes}} \cdots \mathbin{\hat{\otimes}} V \qquad (n \text{ copies}).$$

We put $V^{[1]} = V$ and $V^{[0]} = k$. For any integers $n, m \geqslant 0$, there is a canonical map

$$V^{[n+m]} \to V^{[n]} \mathbin{\hat{\otimes}} V^{[m]}$$

which is identified with the identity unless $n+m=1$, and the canonical map $V \to \hat{V}$ if $n+m=1$. Let

$$C(V) = \prod_{n=0}^{\infty} V^{[n]}$$

with the product topology. By (1.13), we can identify

$$C(V) \hat{\otimes} C(V) = \prod_{n,m=0} V^{[n]} \hat{\otimes} V^{[m]}.$$

Hence there is a unique continuous linear map

$$\Delta_V \colon C(V) \to C(V) \hat{\otimes} C(V)$$

such that $(\mathrm{proj}_n \hat{\otimes} \mathrm{proj}_m) \circ \Delta_V = \mathrm{cano} \circ \mathrm{proj}_{n+m}$ for any integers $n, m \geqslant 0$. It is easy to see $C(V)$ becomes a topological coalgebra with comultiplication Δ_V and counit $\varepsilon_V = \mathrm{proj}_0 \colon C(V) \to k$.

3.1. THEOREM. *Put*

$$\eta = \mathrm{proj}_1 \colon C(V) \to V.$$

For any topological coalgebra C and a continuous linear map $f \colon C \to V$, there is a unique topological coalgebra map $F \colon C \to C(V)$ such that $f = \eta \circ F$.

Proof. Let (Δ, ε) be the coalgebra structure of C. We define continuous linear maps

$$f_n \colon \quad C \to V^{[n]}, \qquad n = 0, 1, 2, \ldots,$$

as follows: $f_0 = \varepsilon$, $f_1 = f$. For $n > 1$, let

$$f_n \colon C \xrightarrow{\Delta_{n-1}} C^{[n]} \xrightarrow{f^{[n]}} V^{[n]},$$

where Δ_{n-1} denotes the $n-1$ times iterated comultiplication. Define a continuous linear map $F \colon C \to C(V)$ by the condition $\mathrm{proj}_n \circ F = f_n$. It is easy to check that F is a unique topological coalgebra map such that $f = \eta \circ F$.

Q.E.D.

By the above universal mapping property, $(C(V), \eta)$ is called the *cofree topological coalgebra* on V. The largest subspace D of $C(V)$ such that $\Delta_V(D) \subset D \otimes D$ gives the cofree coalgebra on V [2, 6.4.1] when V is discrete.

Our next aim is to construct the universal measuring topological coalgebra [2, 7.0.4]. We will define the universal measuring topological coalgebra $\mathrm{Mes}(A, B)$ as a closed subcoalgebra of $C(\mathrm{Hom}(A, B))$ for algebras A, B. In fact, we define it for discrete A and complete B. The notion of complete topological algebras is introduced in the following as dual to topological coalgebras. To use $C(\mathrm{Hom}(A, B))$, it is worth noting that

$$\mathrm{Hom}(M^{[n]}, V^{[n]}) \simeq \mathrm{Hom}(M, V)^{[n]}$$

by (1.14), hence

$$C(\mathrm{Hom}(M, V)) = \prod_{n=0}^{\infty} \mathrm{Hom}(M^{[n]}, V^{[n]})$$

for a discrete vector space M and a topological vector space V.

A *topological algebra* is a topological vector space and algebra B such that the multiplication m_B: $B \otimes B \to B$ is continuous. If B is complete, we have the completion

$$\hat{m}_B\colon B \mathbin{\hat{\otimes}} B \to B.$$

In the following, we fix a complete topological algebra B. If C is a topological coalgebra, then for continuous linear maps $f, g\colon C \to B$, we define

$$f * g\colon C \xrightarrow{\Delta} C \mathbin{\hat{\otimes}} C \xrightarrow{f \hat{\otimes} g} B \mathbin{\hat{\otimes}} B \xrightarrow{\hat{m}_B} B.$$

The continuous linear maps $\mathrm{Hom}_c(C, B)$ form an algebra with multiplication $*$ and unit $u_B \circ \varepsilon$ (u_B the unit for B).

Let A be an algebra. We will construct a closed subcoalgebra $\mathrm{Mes}(A, B)$ of $C(\mathrm{Hom}(A, B))$ in a natural way and show there is a one-to-one correspondence between topological coalgebra maps $C \to \mathrm{Mes}(A, B)$ and algebra maps $A \to \mathrm{Hom}_c(C, B)$ for any topological coalgebra C.

We define a small category $\mathbf{I}$ as follows: The objects are integers 0, 1, 2.... For an integer $n \geqslant 0$, let $[n] = \{1, 2, ..., n\}$. Thus $[0]$ is empty. For integers $n, m \geqslant 0$, $\mathbf{I}(n, m)$ is the set of nondecreasing functions $[n] \to [m]$. We have a category $\mathbf{I}$ with composition obviously defined. Note that $[1]$ is the final object in $\mathbf{I}$, hence there is a unique $\mathbf{I}$-map $[r] \to [1]$ for any $r \geqslant 0$. The category $\mathbf{I}$ has the following $+$-structure: For integers $n, m \geqslant 0$, we put $[n] + [m] = [n+m]$. For two $\mathbf{I}$-maps σ: $[n] \to [n']$ and τ: $[m] \to [m']$, define the $\mathbf{I}$-map $\rho = \sigma + \tau$: $[n+m] \to [n'+m']$ by

$$\rho(i) = \sigma(i) \quad \text{if } i \leqslant n, \qquad \rho(i) = \tau(i-n) + n' \quad \text{if } i > n.$$

We have an associative, but not commutative, functor $+ : \mathbf{I} \times \mathbf{I} \to \mathbf{I}$. The following lemma is trivial:

3.2. LEMMA. *For any* **I**-*map* $\sigma: [n] \to [m]$, *there is a unique decomposition* $n = r_1 + \cdots + r_m$ *such that* $\sigma = \sigma_1 + \cdots + \sigma_m$, *where* $\sigma_i: [r_i] \to [1]$ *is the unique* **I**-*map for* $i = 1, \ldots, m$.

We will associate with the complete algebra B a functor $[B]$ from the category **I** into the category of topological vector spaces as follows: With an integer $n \geqslant 0$, we associate the topological vector space $B^{[n]}$. With the unique **I**-map $\sigma: [r] \to [1]$, we associate $\hat{m}_{r-1}: B^{[r]} \to B$ which is the $r-1$ times iterated multiplication if $r > 1$, the identity I_B if $r = 1$, and the unit u_B if $r = 0$. For a general **I**-map $\tau: [n] \to [m]$, we apply (3.2) to decompose it into the sum $\tau = \tau_1 + \cdots + \tau_m$, where $\tau_i: [r_i] \to [1]$ and $n = r_1 + \cdots + r_m$. We associate with τ the map

$$B^{[n]} = B^{[r_1]} \hat{\otimes} \cdots \hat{\otimes} B^{[r_m]} \xrightarrow{\tau_1 \hat{\otimes} \cdots \hat{\otimes} \tau_m} B \hat{\otimes} \cdots \hat{\otimes} B = B^{[m]},$$

where $\tau_i: B^{[r_i]} \to B$ denotes the map associated with τ_i. We denote this continuous linear map by τ_B. We leave it to the reader to check that we have a functor $[B]$ by the above correspondence.

We note that any algebra is a complete topological algebra with the discrete topology. In such a case, $\hat{\otimes}$ reduces to $\otimes$. Hence if A is an algebra, we can associate a functor $[A]$ from **I** into the category of vector spaces. In the following, we fix a discrete algebra A.

A natural transformation $\alpha: [A] \to [B]$ is a set of linear maps $\alpha = (\alpha_0, \alpha_1, \ldots)$, where $\alpha_n: A^{[n]} \to B^{[n]}$ such that

$$\sigma_B \circ \alpha_n = \alpha_m \circ \sigma_A \tag{3.3}$$

for any **I**-map $\sigma: [n] \to [m]$. The set of all natural transformations $[A] \to [B]$ form a subspace of $\prod_{n=0} \mathrm{Hom}(A^{[n]}, B^{[n]}) = C(\mathrm{Hom}(A, B))$. The subspace is denoted by

$$\mathrm{Mes}(A, B).$$

Since the maps in **I** are generated by injective **I**-maps $[n] \to [n+1]$ and surjective **I**-maps $[n+1] \to [n]$, we may identify the functor $[B]$ as a complex of the form

$$k \to B \overset{\rightarrow}{\underset{\leftarrow}{\rightarrow}} B^{[2]} \overset{\rightarrow}{\underset{\leftarrow}{\overset{\rightarrow}{\underset{\leftarrow}{\rightarrow}}}} B^{[3]} \cdots$$

with arrows corresponding to the above generators. The vector space $\mathrm{Mes}(A, B)$ is precisely the vector space of homomorphisms of the complex of A into the complex of B.

3.4. THEOREM. *The vector space* $\mathrm{Mes}(A, B)$ *is a closed subcoalgebra of* $C(\mathrm{Hom}(A, B))$.

Proof. Put $M = \mathrm{Mes}(A, B)$ and $H = \mathrm{Hom}(A, B)$. It follows from condition (3.3) that we can represent M as the kernel of a continuous linear double map

$$\prod_{n=0}^{\infty} \mathrm{Hom}(A^{[n]}, B^{[n]}) \rightrightarrows \prod_{\sigma:[n] \to [m]} \mathrm{Hom}(A^{[n]}, B^{[m]}), \tag{3.5}$$

where the right term is the product over all **I**-maps. This shows the closedness of M. If $\alpha = (\alpha_0, \alpha_1, ...)$ is in M, we have $\Delta_H(\alpha) = (\alpha_{n+r})_{n,r}$ by definition of the comultiplication Δ_H of $C(H)$. Apply $- \hat{\otimes}\, C(H)$ to (3.5) and get

$$\prod_{n,r=0}^{\infty} \mathrm{Hom}(A^{[n+r]}, B^{[n+r]}) \rightrightarrows \prod_{\sigma:[n] \to [m],r} \mathrm{Hom}(A^{[n+r]}, B^{[m+r]}) \tag{3.6}$$

where the right product is taken over all pairs (σ, r) of **I**-maps σ and **I**-objects r. Here we use (1.13) and (1.14), and identify

$$\mathrm{Hom}(A^{[n]}, B^{[m]}) \hat{\otimes} \mathrm{Hom}(A^{[r]}, B^{[r]}) = \mathrm{Hom}(A^{[n+r]}, B^{[m+r]}).$$

By (2.2), the kernel of (3.6) is $M \hat{\otimes} C(H)$. Now the double map (3.6) sends $(\beta_{n,r})_{n,r}$ to $((\gamma_{\sigma,r})_{\sigma,r}, (\delta_{\sigma,r})_{\sigma,r})$, where $\gamma_{\sigma,r} = (\sigma + i_r)_B \circ \beta_{n,r}$ and $\delta_{\sigma,r} = \beta_{m,r} \circ (\sigma + i_r)_A$ for an **I**-map σ: $[n] \to [m]$ with identity i_r: $[r] \to [r]$. It follows immediately that $\Delta_H(\alpha)$ is in the kernel for $\alpha \in M$. Hence $\Delta_H(M) \subset M \hat{\otimes} C(H)$. Similarly we have $\Delta_H(M) \subset C(H) \hat{\otimes} M$. It follows from an easy diagram-chasing that $M \hat{\otimes} M = M \hat{\otimes} C(H) \cap C(H) \hat{\otimes} M$. Therefore $\Delta_H(M) \subset M \hat{\otimes} M$, and M is a subcoalgebra. Q.E.D.

Let C be a topological coalgebra. There is a one-to-one correspondence between a continuous linear map ϕ: $C \to \mathrm{Hom}(A, B)$ and a linear map ψ: $A \to \mathrm{Hom}_c(C, B)$ by $\phi(c)(a) = \psi(a)(c)$ for $c \in C$, $a \in A$. Define the continuous linear maps ϕ_n: $C \to \mathrm{Hom}(A^{[n]}, B^{[n]})$, $n = 0, 1, ...$ as in the proof of (3.1). Thus (ϕ_n): $C \to C(\mathrm{Hom}(A, B))$ is the topological coalgebra map corresponding to ϕ. The linear map ψ_n: $A^{[n]} \to \mathrm{Hom}_c(C, B^{[n]})$ corresponding to ϕ_n is as follows:

$$\psi_n(a_1 \otimes \cdots \otimes a_n): C \xrightarrow{\Delta_{n-1}} C^{[n]} \xrightarrow{\psi(a_1) \hat{\otimes} \cdots \hat{\otimes} \psi(a_n)} B^{[n]}$$

for $a_1, ..., a_n$ in A, where Δ_{n-1} denotes the $n-1$ times iterated comultiplication of C if $n > 1$, the identity if $n = 1$, and the counit ε if $n = 0$.

3.7. THEOREM. *Let A be an algebra, let B be a complete topological algebra, and let C be a topological coalgebra. Assume a continuous linear map ϕ: $C \to \mathrm{Hom}(A, B)$ corresponds to a linear map ψ: $A \to \mathrm{Hom}_c(C, B)$,* *and let* $\phi_n\colon C \to \mathrm{Hom}(A^{[n]}, B^{[n]})$ *and* $\psi_n\colon A^{[n]} \to \mathrm{Hom}_c(C, B^{[n]})$ *be as above for integers* $n \geqslant 0$. *The following statements are equivalent*:

(i) ψ *is an algebra map.*

(ii) $\mathrm{Hom}_c(I_C, \hat{m}_B) \circ \psi_2 = \psi \circ m_A$ *and* $\mathrm{Hom}_c(I_C, u_B) \circ \psi_0 = \psi \circ u_A$, *where* (m_B, u_B) *and* (m_A, u_A) *denote the algebra structures of* B *and* A.

(iii) $\psi_n\colon A^{[n]} \to \mathrm{Hom}_c(C, B^{[n]})$ $(n \in \mathbf{I})$ *give rise to a natural transformation of functors of* $\mathbf{I}$.

(iv) *The image of the topological coalgebra map* $(\phi_n)\colon C \to C(\mathrm{Hom}(A, B))$ *is in* $\mathrm{Mes}(A, B)$.

Proof. It follows easily that (i)$\Leftrightarrow$(ii) and (iii)$\Leftrightarrow$(iv). (ii) follows from (iii). The set of $\mathbf{I}$-maps $\sigma\colon [n] \to [m]$ such that $\mathrm{Hom}_c(I_C, \sigma_B) \circ \psi_n = \psi_m \circ \sigma_A$ is closed under composition and addition "+." Since the $\mathbf{I}$-maps $[2] \to [1]$ and $[0] \to [1]$ generate all $\mathbf{I}$-maps under composition and addition, (iii) follows from (ii). Q.E.D.

A continuous linear map $\phi\colon C \to \mathrm{Hom}(A, B)$ is called a *measuring* [2, Sect. 7.0] if the above equivalent conditions are satisfied. As a corollary we have:

3.8. THEOREM. *Let* A *be an algebra and let* B *be a complete topological algebra. For any topological coalgebra there is a one-to-one correspondence among*:

ϕ: $C \to \mathrm{Hom}(A, B)$, *a continuous measuring,*

ψ: $A \to \mathrm{Hom}_c(C, B)$, *an algebra map,*

F: $C \to \mathrm{Mes}(A, B)$, *a topological coalgebra map,*

where F *is the topological coalgebra map* $C \to C(\mathrm{Hom}(A, B))$ *corresponding to* ϕ, *and* $\phi(c)(a) = \psi(a)(c)$ *for* $c \in C$, $a \in A$.

In this sense, $\mathrm{Mes}(A, B)$ can be called the *universal measuring topological coalgebra.*

4. THE CO-HOM OBJECT IN TOPOLOGICAL VECTOR SPACES

For two topological vector spaces V and W, denote by $\mathrm{Hom}_c(V, W)$ the vector space of all continuous linear maps $V \to W$. We begin this section by showing that the functor $V \mapsto W \hat{\otimes} V$ is *representable.* More precisely we endow $W^{\odot}$ with some linear topology and show there is a canonical linear isomorphism

$$W \hat{\otimes} V \simeq \mathrm{Hom}_c(W^{\odot}, \hat{V}).$$

If we take $V=k$, this gives $\hat{W} \simeq W^{\odot\odot}$. We show the inverse ω_W: $W^{\odot\odot} \to \hat{W}$ is continuous but not necessarily homeomorphic. Let C be a topological vector space. We obtain the following diagram from the above isomorphism for C instead of V:

$$\begin{aligned}\operatorname{Hom}(V, W \hat{\otimes} C) \simeq \operatorname{Hom}(V, \operatorname{Hom}_c(W^{\odot}, \hat{C})) \subset \operatorname{Hom}(V, \operatorname{Hom}(W^{\odot}, \hat{C}))\\ \simeq \operatorname{Hom}(W^{\odot} \otimes V, \hat{C}).\end{aligned}$$

We will prove there is some special linear topology on $W^{\odot} \otimes V$ called *the Ω-topology* such that if we denote by $W^{\odot} \otimes_{\Omega} V$ the topological vector space with the Ω-topology, then the composite of the above diagram induces a linear isomorphism

$$\operatorname{Hom}_c(V, W \hat{\otimes} C) \simeq \operatorname{Hom}_c(W^{\odot} \otimes_{\Omega} V, \hat{C}).$$

This means we can say $c(W, V) = W^{\odot} \otimes_{\Omega} V$ is the co-hom object in the category of topological vector spaces with respect to $\hat{\otimes}$. It follows that $c(V, V)$ has a natural structure of a topological coalgebra. We call this the *co-end coalgebra* of V. Its continuous dual is naturally isomorphic to the continuous endomorphism algebra $\operatorname{End}_c(\hat{V})$.

We call a topological vector space W is *representable* if there is a topological vector space W' together with a family of linear isomorphisms

$$t_V: W \hat{\otimes} V \simeq \operatorname{Hom}_c(W', \hat{V})$$

which is defined for all topological vector spaces V and natural in V. In this case, we also say W' represents W along $\{t_V\}$. We claim that all topological vector spaces are representable.

4.1. LEMMA. *Let W and W' be topological vector spaces. W' represents W if there is a family of linear isomorphisms*

$$t_M: W \hat{\otimes} M \simeq \operatorname{Hom}_c(W', M)$$

which is defined for all discrete topological vector spaces M and natural in M.

Proof. Let $\{V_\alpha\}$ be a topological basis of the topological vector space V. We have two natural isomorphisms:

$$W \hat{\otimes} V \simeq \varprojlim_{\alpha} W \hat{\otimes} (V/V_\alpha),$$

$$\operatorname{Hom}_c(W', \hat{V}) \simeq \varprojlim_{\alpha} \operatorname{Hom}_c(W', V/V_\alpha)$$

by (2.3) and definition. If we define t_V to be the projective limit of $t_{(V/V_\alpha)}$, then it is easy to see t_V is natural in V. Q.E.D.

4.2 PROPOSITION. *Let N be a discrete topological vector space. Then N^* represents N.*

Proof. Assume N is finite dimensional at first. Then N^* is also discrete and we have a trivial isomorphism

$$N \otimes M \simeq \operatorname{Hom}(N^*, M)$$

for all discrete topological vector spaces M. In general, let $\{N_i\}$ be the set of all finite dimensional subspaces of N. Then $\{N_i^{\perp}\}$ form a topological basis for N^*. By definition we have

$$\operatorname{Hom}_c(N^*, M) = \bigcup_i \operatorname{Hom}(N^*/(N_i^{\perp}), M) = \bigcup_i \operatorname{Hom}(N_i^*, M).$$

It follows from $N_i \otimes M \simeq \operatorname{Hom}(N_i^*, M)$ that we have a natural linear isomorphism

$$N \otimes M \simeq \operatorname{Hom}_c(N^*, M)$$

defined for all discrete topological vector spaces M. Hence N^* represents N by (4.1) above. Q.E.D.

We give the inductive limit topology on the continuous dual $W^{\odot}$ as follows: Let $\{W_\gamma\}$ be a topological basis of W. We know $W^{\odot}$ is the directed union of $(W/W_\gamma)^*$. Give on $W^{\odot}$ the inductive limit topology of the topologies on $(W/W_\gamma)^*$. We have

$$\operatorname{Hom}_c(W^{\odot}, V) = \varprojlim_{\gamma} \operatorname{Hom}_c((W/W_\gamma)^*, V) \tag{4.3.1}$$

for any topological vector space V.

4.3. THEOREM. *For any topological vector space W, the topological vector space $W^{\odot}$ represents W.*

Proof. For any topological vector space V, we have

$$\begin{aligned} W \hat{\otimes} V = \varprojlim_{\gamma} \{(W/W_\gamma) \hat{\otimes} V\} &\simeq \varprojlim_{\gamma} \operatorname{Hom}_c((W/W_\gamma)^*, \hat{V}) \\ &= \operatorname{Hom}_c(W^{\odot}, \hat{V}) \end{aligned}$$

by (2.3), (4.2), and (4.3.1). This means $W^{\odot}$ represents W. Q.E.D.

The canonical isomorphism between $W \hat{\otimes} V$ and $\operatorname{Hom}_c(W^{\odot}, \hat{V})$ is described as follows:

4.4. THEOREM. *Let V and W be topological vector spaces. For z in $W^{\odot}$, we have a continuous linear map*

$$z \hat{\otimes} I\colon \quad W \hat{\otimes} V \to k \hat{\otimes} V = \hat{V}.$$

If u is in $W \hat{\otimes} V$, then the linear map

$$\tau(u)\colon \quad z \mapsto (z \hat{\otimes} I)(u),\ W^{\odot} \to \hat{V}$$

is continuous, and the map $u \mapsto \tau(u)$ gives rise to a linear isomorphism

$$\tau\colon \quad W \hat{\otimes} V \simeq \mathrm{Hom}_c(W^{\odot}, \hat{V})$$

which is natural in W and V.

Proof. By the argument of (4.1) through (4.3), we can assume V and W are discrete and W is finite dimensional as well. In this case the claim is trivially true. Q.E.D.

The following linear isomorphism follows from the above:

$$\hat{W} = W \hat{\otimes} k \overset{\tau}{\simeq} \mathrm{Hom}_c(W^{\odot}, k) = W^{\odot\odot}.$$

Let

$$\omega_W\colon \quad W^{\odot\odot} \to \hat{W}$$

be the inverse of the above linear isomorphism. This is obviously natural in W and satisfies the defining identity:

$$\hat{z}(\omega_W(c)) = c(z) \tag{4.5.1}$$

for z in $W^{\odot}$ and c in $W^{\odot\odot}$.

4.5. PROPOSITION. *For any topological vector space W, the linear isomorphism ω_W: $W^{\odot\odot} \to \hat{W}$ is continuous.*

Proof. By the naturality of ω_W, we can reduce to the case $W = N$ is discrete. It is enough to show $N^{*\odot}$ is discrete. With the notation of (4.2), $N^{*\odot}$ is the inductive limit of the discretre spaces N_i^*, hence discrete itself. Q.E.D.

The map ω_W is a homeomorphism if W is discrete or more generally a direct product of discrete spaces as we prove below. Hence if W is $\mathrm{Hom}(M, N)$ or M^* with discrete M and N, then ω_W is a homeomorphism. We give below an example of W whose ω_W is *not* a homeomorphism.

4.6.1. EXAMPLE. Let W be a vector space with a linear basis Λ whose cardinality is larger than the countable infinity. We identify W^* with the set of all functions $\Lambda \to k$. Give the following linear topology on W: A subspace L of W is open if the cardinality of the set of elements $\lambda \in \Lambda$ which do not belong to L is smaller than the cardinality of Λ. We claim $W^{\odot\odot}$ is discrete. Since $\hat{W}$ which contains the nondiscrete subspace W is not discrete, it follows that *ω_W is not a homeomorphism.*

We introduce the following notation for the proof. Let k^S be the set of all functions of the set S into k. Thus $W^* = k^\Lambda$. If $T \subset S$, we identify k^T with the subspace of all f in k^S which vanishes on $S - T$. Call a subset P of Λ proper if its cardinality is smaller than the cardinality of Λ. Then $W^\odot$ is the union of k^P for all proper subsets P of Λ. We claim the topology of $W^\odot$ is induced from $W^* = k^\Lambda$ (the direct product of copies of k). If this is true, all open subspaces of $W^\odot$ are cofinite. Hence it follows that $W^{\odot\odot}$ is discrete. To prove the claim, let U be an open subspace of $W^\odot$ which is not open relative to the induced topology from W^*. For any proper subset P of Λ, the intersection $U \cap k^P$ is open in k^P, hence there is a finite subset F of P such that U contains k^{P-F}. There is the smallest subset F_P (relative to the inclusion) among such finite subsets F since those finite subsets are closed relative to the intersection. We will construct a sequence of finite subsets $\{F_0, F_1, \ldots\}$ and a sequence of proper subsets $\{P_0, P_1, \ldots\}$ of Λ by induction as follows: F_0 is arbitrary. Since U does not contain $W^\odot \cap k^{\Lambda - F_0}$, there is a proper subset P_0 of $\Lambda - F_0$ such that $k^{P_0} \not\subset U$. Assume we already have a sequence of finite subsets $F_0 \subset F_1 \subset \cdots \subset F_k$. Let P_k be a proper subset of $\Lambda - F_k$ such that $k^{P_k} \not\subset U$. Let F_{k+1} be the (disjoint) union of F_k with $F_{(P_k)}$. Thus we obtain two sequences $\{F_i\}$ and $\{P_i\}$. Let Q be the union of all P_i, $0 \leqslant i < \infty$. We have a finite subset F_Q since Q is proper. For all $i \geqslant 0$, the set $F_{(P_i)}$ is contained in F_Q since we have $k^{P_i - F_Q} \subset k^{Q - F_Q} \subset U$. But since the family of finite sets $F_{(P_i)}$, $i = 0, 1, \ldots$, is disjoint and consists of nonempty sets, this contradicts to the finiteness of F_Q. It follows that the topology of $W^\odot$ is induced from W^*.

4.6.2. EXAMPLE. Let W be the direct product of discrete topological vector spaces $N^{(p)}$, $p \in I$. Then ω_W is a homeomorphism.

Indeed $W^\odot$ is the direct sum of $N^{(p)*}$, $p \in I$. It follows from the definition of the topology of $W^\odot$ and the commutativity of inductive limits that the topology on $W^\odot$ is the direct sum topology. This means if $\mathscr{J}$ denotes the set of all families $L = \{L^{(p)}\}_{p \in I}$, where $L^{(p)}$ is a finite dimensional subspace of $N^{(p)}$, then $W^\odot$ has a topological basis $\{\bigoplus_p L^{(p)\perp}\}_{L \in \mathscr{J}}$. Hence by definition, $W^{\odot\odot}$ is the inductive limit of the direct products $\prod_p L^{(p)}$, $L \in \mathscr{J}$ as a topological vector space. All we have to prove is show that this topological vector space is identical to the direct product $W = \prod_p N^{(p)}$. This is done by the diagonal argument as follows. We assume I is infinite.

Let U be a subspace of W which is not open relative to the product topology but open relative to the inductive limit topology. This means for any finite subset F of I, the product $\prod_{p \in I-F} N^{(p)}$ is never contained in U, but the intersection $U \cap \prod_p L^{(p)}$ is an open subspace of $\prod_p L^{(p)}$ for all L in J. We construct a sequence of elements $\lambda_1, \lambda_2, \ldots,$ in I and a sequence of elements $f_1, f_2, \ldots,$ in $W-U$ by induction as follows: The element λ_1 is arbitrary. Let f_1 be an element in $W-U$ such that $f_1(\lambda_1)=0$. Assume we already have elements $\lambda_1, \lambda_2, \ldots, \lambda_k$ and $f_1, f_2, \ldots, f_k$ such that $f_i(\lambda_j)=0$ for $1 \leqslant j \leqslant i \leqslant k$, but $f_i(\lambda_{i+1}) \neq 0$ for $1 \leqslant i < k$. Let λ_{k+1} be an element in I such that $f_k(\lambda_{k+1}) \neq 0$. There is an element f_{k+1} in $W-U$ which vanishes on $\lambda_1, \lambda_2, \ldots, \lambda_{k+1}$. Thus we obtain sequences $\{\lambda_i\}$ and $\{f_i\}$. For $j \geqslant 1$, let $L^{(j)}$ be the subspace of $N^{(\lambda_j)}$ spanned by all elements $f_i(\lambda_j)$, $i \geqslant 1$. This is finite dimensional since $f_i(\lambda_j)=0$ for $i \geqslant j$. For any $i \geqslant 1$, the element f_i belongs to the product $\prod_{k=i+1}^{\infty} L^{(k)}$ but not in U. This contradicts to the assumption that the intersection $U \cap \prod_{k=1}^{\infty} L^{(k)}$ is open in $\prod_{k=1}^{\infty} L^{(k)}$.

For two topological vector spaces V, W, consider the following natural linear isomorphism induced from τ:

$$\operatorname{Hom}_c(W^{\odot}, \hat{V}) \overset{\tau^{-1}}{\simeq} W \hat{\otimes} V \overset{\text{transpose}}{\simeq} V \hat{\otimes} W \overset{\tau}{\simeq} \operatorname{Hom}_c(V^{\odot}, \hat{W}). \quad (4.7.1)$$

We describe the map by means of ω_W or ω_V.

4.7. THEOREM. *Let V, W be topological vector spaces. Let $f\colon W^{\odot} \to \hat{V}$ and $g\colon V^{\odot} \to \hat{W}$ be continuous linear maps. The following are equivalent*:

(a) *f and g correspond to each other under the map of* (4.7.1).

(b) $g = \omega_W \circ f^{\odot}\colon V^{\odot} (= \hat{V}^{\odot}) \to W^{\odot\odot} \to \hat{W}$.

(c) $f = \omega_V \circ g^{\odot}\colon W^{\odot} (= \hat{W}^{\odot}) \to V^{\odot\odot} \to \hat{V}$.

Proof. (a) $\Rightarrow$ (b). Let $f = \tau(u)$, $g = \tau(v)$ with u in $W \hat{\otimes} V$, v in $V \hat{\otimes} W$. For any x in $V^{\odot}$ and z in $W^{\odot}$, we have

$$\begin{aligned}\hat{z}(g(x)) &= \hat{z}((x \hat{\otimes} I)(v)) = (x \hat{\otimes} z)(v) = (z \hat{\otimes} x)(u) \\ &= \hat{x}((z \hat{\otimes} I)(u)) = \hat{x}(f(z)) = f^{\odot}(\hat{x})(z) = \hat{z}(\omega_W(f^{\odot}(x))).\end{aligned}$$

Since the elements in $\hat{W}$ operate faithfully on $W^{\odot}$ as we see above (4.5), it follows that $g(x) = \omega_W(f^{\odot}(x))$. We have (a) $\Rightarrow$ (c) by symmetry. Q.E.D.

4.8. COROLLARY. *For any topological vector space W, the composite*

$$W^{\odot} \xrightarrow{(\omega_W)^{\odot}} W^{\odot\odot\odot} \xrightarrow{\omega_{(W^{\odot})}} \widehat{W^{\odot}}$$

is the canonical map.

Proof. If we put $V = W^{\odot}$ in (4.7.1), then the canonical map $W^{\odot} \to \hat{V}$ corresponds to ω_W: $V^{\odot} \to \hat{W}$. Q.E.D.

Now we proceed into constructing the co-hom object. Let V, W, and C be topological vector spaces. We define the δ-map as the composite:

$$\begin{aligned} \delta\colon \operatorname{Hom}(V, W \hat{\otimes} C) \xrightarrow{\operatorname{Hom}(I,\tau)} & \operatorname{Hom}(V, \operatorname{Hom}(W^{\odot}, \hat{C})) \\ & \simeq \operatorname{Hom}(W^{\odot} \otimes V, \hat{C}). \end{aligned} \tag{4.9.1}$$

For a linear map $f\colon V \to W \hat{\otimes} C$, we write $\delta_f\colon W^{\odot} \otimes V \to \hat{C}$ instead of $\delta(f)$. We have $\delta_f(z \hat{\otimes} v) = (z \hat{\otimes} I)(f(v))$ for z in $W^{\odot}$ and v in V.

4.9. Definition. Let V and W be topological vector spaces. A linear topology on $W^{\odot} \otimes V$ is called an *Ω-topology* if the δ-map induces a linear isomorphism

$$\delta\colon \operatorname{Hom}_c(V, W \hat{\otimes} C) \simeq \operatorname{Hom}_c(W^{\odot} \otimes_{\Omega} V, \hat{C}) \tag{4.9.2}$$

for any topological vector space C, where $W^{\odot} \otimes_{\Omega} V$ denotes the topological vector space with the Ω-topology.

We will prove the existence and uniqueness of Ω-topology. The defining isomorphism (4.9.2) means that the completion of the functor $- \otimes_{\Omega} -$ gives rise to the co-hom object in the category of all *complete* topological vector spaces with product $\hat{\otimes}$. We regard $W^{\odot} \otimes_{\Omega} V$ itself as the co-hom object and put it $c(W, V)$ since our viewpoint is to consider two topological vector spaces are almost equivalent if their completions are isomorphic.

The next lemma shows the uniqueness and functoriality of the Ω-topology.

4.10. Lemma. (a) *Let $\mathscr{S}$ and $\mathscr{T}$ be two linear topologies on the same vector space V, and let $V_{\mathscr{S}}$, $V_{\mathscr{T}}$ denote the topological vector spaces with $\mathscr{S}$, $\mathscr{T}$, respectively. If* $\operatorname{Hom}_c(V_{\mathscr{S}}, N) = \operatorname{Hom}_c(V_{\mathscr{T}}, N)$ *for all discrete vector spaces N, then $\mathscr{S} = \mathscr{T}$.*

(b) *Let W and W' be topological vector spaces and let $f\colon W \to W'$ be a linear map. If the map* $\operatorname{Hom}(f, I)\colon \operatorname{Hom}(W', N) \to \operatorname{Hom}(W, N)$ *maps* $\operatorname{Hom}_c(W', N)$ *into* $\operatorname{Hom}_c(W, N)$ *for all discrete vector spaces N, then f is continuous.*

Proof. (a) follows from applying (b) to the identity $I\colon V \to V$. To prove (b), let N be the quotient of W' by an open subspace. The projection $W' \to N$ composed with f being continuous, the inverse image of the open subspace is open in W. Q.E.D.

By the Ω-topology being functorial we mean that if $f\colon V \to V'$ and $g\colon W' \to W$ are continuous linear maps of topological vector spaces, then the map $g^{\odot} \otimes f\colon W^{\odot} \otimes_{\Omega} V \to W'^{\odot} \otimes_{\Omega} V'$ is continuous. This functoriality will be used to prove the existence.

4.11. THEOREM. *The Ω-topology on $W^{\odot} \otimes V$ exists for any topological vector spaces V, W.*

Proof. First, we reduce to the case W is discrete. Assume there is an Ω-topology on $M^* \otimes V$ for any discrete M. Let $\{W_\gamma\}$ be a topological basis of W. Define $W^{\odot} \otimes_{\Omega} V$ to be the inductive limit of topological vector spaces $(W/W_\gamma)^* \otimes_{\Omega} V$. Since then $\mathrm{Hom}_c(W^{\odot} \otimes_{\Omega} V, \hat{C})$ is the projective limit of the vector spaces $\mathrm{Hom}_c((W/W_\gamma)^* \otimes_{\Omega} V, \hat{C})$, it follows easily that this topology yields the isomorphism (4.9.2).

Let M be a discrete vector space and let $\{V_\alpha\}$ be a topological basis of V. We define a subspace U of $M^* \otimes V$ open if U contains $M^* \otimes V_\alpha$ for some α and if the subspace $\{f \text{ in } M^* \mid f \otimes v \text{ is in } U\}$ is open in M^* for every v in V. This topology is finer than the tensor product topology. Let $M^* \underline{\otimes} V$ be this topological vector space. We claim this is the Ω-topology. We note that the isomorphism (4.9.2) holds if only it does for all discrete vector spaces C since both sides of (4.9.2) preserves projective limits as functors of C. Let N be a discrete vector space. We have

$$\mathrm{Hom}_c(M^* \underline{\otimes} V, N) = \varinjlim_{\alpha} \mathrm{Hom}_c(M^* \underline{\otimes} (V/V_\alpha), N)$$

by definition of the $\underline{\otimes}$-topology, and

$$\mathrm{Hom}_c(V, M \otimes N) = \varinjlim_{\alpha} \mathrm{Hom}(V/V_\alpha, M \otimes N)$$

since $M \otimes N$ is discrete. It follows that we can reduce to the case V is also discrete. What we have to prove is show

$$\delta\colon \mathrm{Hom}(L, M \otimes N) \simeq \mathrm{Hom}_c(M^* \underline{\otimes} L, N)$$

for all discrete vector spaces L, M, and N. But $M^* \underline{\otimes} L$ is the direct sum of $\dim L$ copies of the topological vector spaces M^*, hence we have

$$\mathrm{Hom}(L, \mathrm{Hom}_c(M^*, N)) \simeq \mathrm{Hom}_c(M^* \underline{\otimes} L, N).$$

Since we have $\tau\colon M \otimes N \simeq \mathrm{Hom}_c(M^*, N)$, the desired isomorphism follows from definition. Q.E.D.

4.12. *Conclusion*

Let V and W be topological vector spaces. We put

$$c(W, V) = W^{\odot} \otimes_{\Omega} V$$

and call it the *co-hom topological vector space* of V into W. This is a functor covariant in V and contravariant in W. For any topological vector space C, the δ-map induces an isomorphism

$$\delta\colon \operatorname{Hom}_c(V, W \hat{\otimes} C) \simeq \operatorname{Hom}_c(c(W, V), \hat{C}) \tag{4.12.1}$$

which is natural in each variable. If we put $C = c(W, V)$, we get a canonical continuous linear map

$$\rho_V^W\colon V \to W \hat{\otimes} c(W, V) \tag{4.12.2}$$

whose δ is the canonical map of $c(W, V)$ into its completion. Since the inverse of (4.12.1) is given by $g \mapsto (I \hat{\otimes} g) \circ \rho_V^W$, it follows that the map ρ_V^W satisfies the following universal mapping property:

(4.12.3) *For any continuous linear map $f\colon V \to W \hat{\otimes} C$, there is a unique continuous linear map $g\colon c(W, V) \to \hat{C}$ such that $f = (I \hat{\otimes} g) \circ \rho_V^W$.*

If we put $C = k$ in (4.12.1), we have $\delta\colon \operatorname{Hom}_c(V, \hat{W}) \simeq c(W, V)^{\odot}$. The inverse is given by $z \mapsto (I \hat{\otimes} z) \circ \rho_V^W$. It will be better to regard this as

$$c(W, V)^{\odot} \simeq \operatorname{Hom}_c(\hat{V}, \hat{W}) \text{ by } z \mapsto (I \hat{\otimes} z) \circ \hat{\rho} \qquad (\text{where } \rho = \rho_V^W) \tag{4.12.4}$$

to reobtain the composite from the cocomposite as we see later. This is true since $\operatorname{Hom}_c(\hat{V}, \hat{W}) = \operatorname{Hom}_c(V, \hat{W})$. In other words, $c(W, V)$ is pre-dual to $\operatorname{Hom}_c(\hat{V}, \hat{W})$. The element ε_V in $c(V, V)^{\odot}$ corresponding to the identity of $\hat{V}$ is called the *counit*. It is easy to see $\varepsilon_V(z \otimes v) = z(v)$ for z in $V^{\odot}$ and v in V.

If v is in V, we have a continuous linear map

$$r_v = - \otimes v\colon W \otimes W^{\odot} \to W \otimes (W^{\odot} \otimes_{\Omega} V)$$

which induces $\hat{r}_v\colon W \hat{\otimes} W^{\odot} \to W \hat{\otimes} c(W, V)$. Let j_W be the element in $W \hat{\otimes} W^{\odot}$ whose τ is the canonical map of $W^{\odot}$ into its completion. Then we have

$$\rho_V^W(v) = \hat{r}_v(j_W) \qquad \text{for any} \quad v \text{ in } V. \tag{4.12.5}$$

For the proof, it is enough to compute the δ of the linear map $v \mapsto \hat{r}_v(j_W)$. It is easy and left to the reader to verify that it is precisely the canonical map of $c(W, V)$ into its completion.

The *cocomposite* map follows from the universal mapping property of ρ_V^W. Let V, W, X be topological vector spaces. There is a unique continuous linear map

$$\Delta\colon c(X, V) \to c(X, W) \hat{\otimes} c(W, V)$$

such that the composite

$$(\rho_W^X \hat{\otimes} I) \circ \rho_V^W \colon V \to W \hat{\otimes} c(W, V) \to X \hat{\otimes} c(X, W) \hat{\otimes} c(X, V)$$

is equal to $(I \hat{\otimes} \Delta) \circ \rho_V^X$. This is called the cocomposite map. The cocomposite is obviously coassociative and the counit we previously defined is actually a counit relative to the cocomposite.

In particular it follows that $c(V, V)$ is a topological coalgebra with comultiplication the cocomposite, and V is a right topological $c(V, V)$-comodule with structure ρ_V^V. Hence $\hat{V}$ is a continuous left $c(V, V)^{\odot}$-module, and the corresponding representation

$$c(V, V)^{\odot} \to \mathrm{End}_c(\hat{V})$$

is an isomorphism by (4.12.4). In this sense, the topological coalgebra $c(V, V)$ is pre-dual to the algebra of continuous linear endomorphisms of $\hat{V}$. It is called the *co-end topological coalgebra* for V.

More generally, the cocomposite is pre-dual to the composite. To specify this, note that there is a canonical injective linear map $V^{\odot} \otimes W^{\odot} \to (V \hat{\otimes} W)^{\odot}$ which sends $f \otimes g$ to $f \hat{\otimes} g$. If V, W, X are topological vector spaces, the composite

$$c(X, W)^{\odot} \otimes c(W, V)^{\odot} \xrightarrow{\text{cano}} (c(X, W) \hat{\otimes} c(W, V))^{\odot} \xrightarrow{\Delta^{\odot}} c(X, V)^{\odot}$$

is identified with the composite map $\mathrm{Hom}_c(\hat{W}, \hat{X}) \otimes \mathrm{Hom}_c(\hat{V}, \hat{W}) \to \mathrm{Hom}_c(\hat{V}, \hat{X})$ through the isomorphism of (4.12.4).

5. Ω-Topological Algebras

We have proved there is a canonical linear isomorphism

$$\mathrm{Hom}_c(V, W \hat{\otimes} C) \simeq \mathrm{Hom}_c(W^{\odot} \otimes_{\Omega} V, \hat{C})$$

for topological vector spaces V, W, and C. Since $W \hat{\otimes} C \simeq C \hat{\otimes} W$, we also have

$$\mathrm{Hom}_c(V, W \hat{\otimes} C) \simeq \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, \hat{W}).$$

Let C be a topological coalgebra. Then $C^{\odot}$ is an algebra as we see in Section 2. We prove that the multiplication $C^{\odot} \otimes_{\Omega} C^{\odot} \to C^{\odot}$ is a continuous map, and the transpose $C^{\odot} \otimes_{\Omega} C^{\odot} \to C^{\odot} \otimes_{\Omega} C^{\odot}$ is a homeomorphism. In other words, the two possible Ω-topologies on $C^{\odot} \otimes C^{\odot}$ are the same. We call $C^{\odot}$ the *Ω-topological algebra* associated with C in the sense that the multiplication is continuous relative to the Ω-topology. If we put $V=W$, the above isomorphism yields

$$\mathrm{Hom}_c(V, V \hat{\otimes} C) \simeq \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, \hat{V}).$$

We prove if V is complete, this yields a one-to-one correspondence between right topological comodule structures $V \to V \hat{\otimes} C$ and left continuous module structures $C^{\odot} \otimes_{\Omega} V \to V$. This means the category of complete right topological C-comodules is isomorphic to the category of complete left Ω-topological $C^{\odot}$-modules. Moreover, we show if V is a right (or left) topological C-comodule, then $V^{\odot}$ is a right (or left) Ω-topological $C^{\odot}$-module and a left (or right) topological C-comodule.

Let V, W, and C be topological vector spaces. If $f\colon V \to W \hat{\otimes} C$ is a continuous linear map, put

$$\sigma(f)(x \otimes v) = (I \hat{\otimes} x) f(v) \in \hat{W}$$

for x in $C^{\odot}$ and v in V. It follows from (4.9) and (4.11) that $f \mapsto \sigma(f)$ gives an isomorphism

$$\sigma\colon \mathrm{Hom}_c(V, W \hat{\otimes} C) \simeq \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, \hat{W}).$$

We call a linear map $u\colon V \to W$ is a *pre-isomorphism* if it is continuous and induces an isomorphism of topological vector spaces $\hat{u}\colon \hat{V} \simeq \hat{W}$, or equivalently if it induces a bijection $\mathrm{Hom}_c(W, N) \to \mathrm{Hom}_c(V, N)$ for any complete (or sufficiently, discrete) topological vector space N.

5.1. LEMMA. *For topological vector spaces C, D, and V, the injective map*

$$C^{\odot} \otimes_{\Omega} (D^{\odot} \otimes_{\Omega} V) \to (C \hat{\otimes} D)^{\odot} \otimes_{\Omega} V,\ x \otimes (y \otimes v) \mapsto (x \hat{\otimes} y) \otimes v$$

is a pre-isomorphism.

Proof. Consider the three σ-maps for C, D, and $C \hat{\otimes} D$:

$$\mathrm{Hom}_c(V, W \hat{\otimes} C \hat{\otimes} D) \overset{\sigma_D}{\cong} \mathrm{Hom}_c(D^{\odot} \otimes_{\Omega} V, W \hat{\otimes} C)$$

$$\overset{\sigma_C}{\cong} \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} (D^{\odot} \otimes_{\Omega} V), \hat{W}),$$

$$\mathrm{Hom}_c(V, W \hat{\otimes} C \hat{\otimes} D) \overset{\sigma_{C \hat{\otimes} D}}{\cong} \mathrm{Hom}_c((C \hat{\otimes} D)^{\odot} \otimes_{\Omega} V, \hat{W})$$

for a topological vector space W. A simple calculation shows that the resulting isomorphism $\mathrm{Hom}_c((C \hat{\otimes} D)^{\odot} \otimes_{\Omega} V, \hat{W}) \simeq \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} (D^{\odot} \otimes_{\Omega} V), \hat{W})$ is induced from the linear map in the statement. Hence the map in question is a pre-isomorphism. Q.E.D.

Similarly we have:

5.2. Lemma. *The map*

$$C^{\odot} \otimes_{\Omega} (D^{\odot} \otimes_{\Omega} V) \to D^{\odot} \otimes_{\Omega} (C^{\odot} \otimes_{\Omega} V),\ x \otimes (y \otimes v) \mapsto y \otimes (x \otimes v)$$

is an isomorphism of topological vector spaces.

This means the two possible Ω-topologies on $C^{\odot} \otimes D^{\odot}$ are the same and the topological vector space $C^{\odot} \otimes_{\Omega} D^{\odot}$ is pre-isomorphic to $(C \hat{\otimes} D)^{\odot}$. We can show the first fact directly: If M and N are discrete vector spaces, then the Ω-topology on $M^* \otimes N^*$ reduces to the tensor product topology as follows from the description in the proof of (4.11). Since the functor $(-)^{\odot} \otimes_{\Omega} (-)^{\odot}$ preserves direct limits, we have

$$C^{\odot} \otimes_{\Omega} D^{\odot} \simeq \varinjlim_{\alpha,\beta} (C/C_{\alpha})^* \otimes (D/D_{\beta})^*$$

if C and D have topological bases $\{C_{\alpha}\}$ and $\{D_{\beta}\}$. It follows immediately that the transpose gives an isomorphism of topological vector spaces $C^{\odot} \otimes_{\Omega} D^{\odot} \simeq D^{\odot} \otimes_{\Omega} C^{\odot}$.

Recall that $C^{\odot}$ is an algebra (Sect. 2) if C is a topological coalgebra.

5.3. Proposition. *Let C be a topological coalgebra. The multiplication $C^{\odot} \otimes_{\Omega} C^{\odot} \to C^{\odot}$ is continuous.*

Proof. The multiplication is the composite of the pre-isomorphism $C^{\odot} \otimes_{\Omega} C^{\odot} \to (C \hat{\otimes} C)^{\odot}$ with $\Delta^{\odot}\colon (C \hat{\otimes} C)^{\odot} \to C^{\odot}$. Q.E.D.

We call $C^{\odot}$ the *Ω-topological algebra* associated with C in the sense that the multiplication is continuous relative to the Ω-topology. A left $C^{\odot}$-module and topological vector space V is called an *Ω-topological module* if the multiplication $C^{\odot} \otimes_{\Omega} V \to V$ is continuous.

5.4. Theorem. *Let C be a topological coalgebra and let V be a complete topological vector space. Under the isomorphism*

$$\begin{array}{ccc} \sigma\colon \mathrm{Hom}_c(V, V \hat{\otimes} C) & \simeq & \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, V) \\ \rho & \leftrightarrow & \sigma(\rho) \end{array}$$

ρ is a right comodule structure if and only if $\sigma(\rho)$ is a left module structure.

Proof. Let $\rho_1 = (I \hat{\otimes} \Delta) \circ \rho$, $\rho_2 = (\rho \hat{\otimes} I) \circ \rho$:

$$V \to V \hat{\otimes} C \to V \hat{\otimes} C \hat{\otimes} C.$$

Then we have

$$\sigma(\sigma(\rho_1)) = \sigma(\rho) \circ (\Delta^{\odot} \otimes I) \circ \text{cano}:$$

$$C^{\odot} \otimes_{\Omega} (C^{\odot} \otimes_{\Omega} V) \to (C \hat{\otimes} C)^{\odot} \otimes_{\Omega} V \to C^{\odot} \otimes_{\Omega} V \to V$$

with the canonical map defined in (5.1). If we denote by μ the multiplication of $C^{\odot}$, this yields

$$\sigma(\sigma(\rho_1)) = \sigma(\rho) \circ (\mu \otimes I): C^{\odot} \otimes_{\Omega} (C^{\odot} \otimes_{\Omega} V) \to C^{\odot} \otimes_{\Omega} V \to V.$$

On the other hand we have

$$\sigma(\sigma(\rho_2)) = \sigma(\rho) \circ (I \otimes \sigma(\rho)): C^{\odot} \otimes_{\Omega} (C^{\odot} \otimes_{\Omega} V) \to C^{\odot} \otimes_{\Omega} V \to V.$$

It follows that ρ is coassociative if and only if $\sigma(\rho)$ is associative. The counit condition is equivalent to the unit condition. Q.E.D.

Let us see what will happen if we drop the assumption of completeness. If N is a complete topological vector space, we have $\operatorname{Hom}_c(V, N) = \operatorname{Hom}_c(\hat{V}, N)$, and $V \hat{\otimes} C = \hat{V} \hat{\otimes} C$. Hence we have

$$\operatorname{Hom}_c(V, V \hat{\otimes} C) = \operatorname{Hom}_c(\hat{V}, \hat{V} \hat{\otimes} C).$$

It is easy to see under this identification we have a one-to-one correspondence between right comodule structures $V \to V \hat{\otimes} C$ and $\hat{V} \to \hat{V} \hat{\otimes} C$. Therefore we have:

5.5. Corollary. *Let C be a topological coalgebra and let V be a topological vector space. Under the isomorphism*

$$\begin{array}{ccc} \operatorname{Hom}_c(V, V \hat{\otimes} C) & \simeq & \operatorname{Hom}_c(C^{\odot} \otimes_{\Omega} \hat{V}, \hat{V}) \\ \rho & \leftrightarrow & \sigma(\rho) \end{array}$$

ρ is a right comodule structure if and only if $\sigma(\rho)$ is a left module structure.

5.6. Lemma. *If $u: V \to W$ is a pre-isomorphism of topological vector spaces, then so is $I \otimes u: C^{\odot} \otimes_{\Omega} V \to C^{\odot} \otimes_{\Omega} W$ for any topological vector space C.*

Proof. Apply $\operatorname{Hom}_c(-, N)$ with complete vector space N. We get the isomorphism $\operatorname{Hom}_c(u, I): \operatorname{Hom}_c(W, N \hat{\otimes} C) \to \operatorname{Hom}_c(V, N \hat{\otimes} C)$. Q.E.D.

This means that the canonical map $V \to \hat{V}$ induces a pre-isomorphism $C^{\odot} \otimes_{\Omega} V \to C^{\odot} \otimes_{\Omega} \hat{V}$. Let V be an Ω-topological left $C^{\odot}$-module with a topological coalgebra C. The multiplication $C^{\odot} \otimes_{\Omega} V \to V$ induces a continuous linear map $C^{\odot} \otimes_{\Omega} \hat{V} \to \hat{V}$. As easily checked, $\hat{V}$ becomes a left

Ω-topological $C^{\odot}$-module. By the previous corollary, V has a structure of a right topological C-comodule. Hence we have:

5.7. COROLLARY. *If C is a topological coalgebra, any Ω-topological left (or right) $C^{\odot}$-module is a topological right (or left) C-comodule.*

In other words, the composite

$$\mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, V) \xrightarrow{\text{cano}} \mathrm{Hom}_c(C^{\odot} \otimes_{\Omega} V, \hat{V}) \simeq \mathrm{Hom}_c(V, V \hat{\otimes} C)$$

maps left module structures into right comodule structures.

Let C be a topological coalgebra and V a right topological C-comodule with structure $\rho: V \to V \hat{\otimes} C$. For x in $V^{\odot}$ and a in $C^{\odot}$, define the right multiplication $x \cdot a$ in $V^{\odot}$ as the composite:

$$x \cdot a = (x \hat{\otimes} a) \circ \rho: V \to V \hat{\otimes} C \to k.$$

Then $V^{\odot}$ is a right $C^{\odot}$-module. Since the multiplication $x \otimes a \mapsto x \cdot a$ is the composite of the canonical pre-isomorphism $V^{\odot} \otimes_{\Omega} C^{\odot} \to (V \hat{\otimes} C)^{\odot}$ with $\rho^{\odot}$, we have:

5.8. PROPOSITION. *Let C be a topological coalgebra. If V is a right (or left) topological C-comodule, then $V^{\odot}$ is a right (or left) Ω-topological $C^{\odot}$-module.*

It follows in view of (5.7) that if V is a right (or left) topological C-comodule, then $V^{\odot}$ is a left (or right) topological C-comodule. Let's describe the structure of $V^{\odot}$ by means of the structure of V directly in the following.

5.9. DEFINITION. Let V, W, and C be topological vector spaces. For a continuous linear map $\rho: V \to W \hat{\otimes} C$, define a continuous linear map $\rho^{\circ}: W^{\odot} \to C \hat{\otimes} V^{\odot}$ as follows: Consider the following composite of continuous linear maps

$$V^{\odot\odot} \otimes_{\Omega} W^{\odot} \overset{\text{transpose}}{\simeq} W^{\odot} \otimes_{\Omega} V^{\odot\odot} \xrightarrow{I \otimes \omega_V} W^{\odot} \otimes_{\Omega} \hat{V} \xrightarrow{\hat{\delta}_{\rho}} \hat{C},$$

where ω_V is the map of (4.5) and the completion of δ_ρ is defined on $W^{\odot} \otimes_{\Omega} \hat{V}$ by (5.6). Let ρ° be the unique continuous linear map such that $\sigma(\rho^{\circ})$ is the above composite. In other words,

$$\delta_{\rho^{\circ}} = \hat{\delta}_{\rho} \circ (I \otimes \omega_V).$$

By the definition of δ-maps, this means that ρ° is the unique continuous linear map such that the composite

$$W^{\odot} \xrightarrow{\rho^\circ} C \hat{\otimes} V^{\odot} \overset{\mathrm{tr}}{\cong} V^{\odot} \hat{\otimes} C \overset{\tau}{\cong} \mathrm{Hom}_c(V^{\odot\odot}, \hat{C})$$

and the composite

$$V^{\odot\odot} \xrightarrow{\omega_V} \hat{V} \xrightarrow{\hat{\rho}} W \hat{\otimes} C \overset{\tau}{\cong} \mathrm{Hom}_c(W^{\odot}, \hat{C})$$

correspond to the same linear map $W^{\odot} \otimes V^{\odot\odot} \to \hat{C}$.

5.10. *Properties of the Correspondence* $\rho \mapsto \rho^\circ$

(a) The linear map $\rho \mapsto \rho^\circ$, $\mathrm{Hom}_c(V, W \hat{\otimes} C) \to \mathrm{Hom}_c(W^{\odot}, C \hat{\otimes} V^{\odot})$ is natural in each variable.

(b) If $C = k$, then for a continuous linear map $f\colon V \to \hat{W}$, the map $f^\circ\colon W^{\odot} \to \widehat{V^{\odot}}$ is the composite of $f^{\odot}$ with the canonical map of $V^{\odot}$ into its completion.

(c) For a in $C^{\odot}$ and ρ in $\mathrm{Hom}_c(V, W \hat{\otimes} C)$, let

$$\rho_a = (I \hat{\otimes} a) \circ \rho\colon V \to W \hat{\otimes} C \to \hat{W},$$
$$(\rho^\circ)_a = (a \hat{\otimes} I) \circ \rho^\circ\colon W^{\odot} \to C \hat{\otimes} V^{\odot} \to \widehat{V^{\odot}}.$$

Then we have $(\rho^\circ)_a = (\rho_a)^\circ$, namely $(\rho_a)^{\odot}$ composed with the canonical map into the completion.

(d) Let U, V, W, C, D be topological vector spaces. Let $\rho_1\colon V \to W \hat{\otimes} C$ and $\rho_2\colon U \to V \hat{\otimes} D$ be continuous linear maps. If ρ is the composite

$$\rho = (\rho_1 \hat{\otimes} I) \circ \rho_2\colon U \to V \hat{\otimes} D \to W \hat{\otimes} (C \hat{\otimes} D))$$

then ρ° is the composite

$$\rho^\circ = (I \hat{\otimes} (\rho_2)^\circ) \circ (\rho_1)^\circ\colon W^{\odot} \to C \hat{\otimes} V^{\odot} \to (C \hat{\otimes} D) \hat{\otimes} U^{\odot}.$$

Proof. (a) and (b) follow from the definition. (c) follows from (a) and (b). To prove (d), note the following injective map

$$C \hat{\otimes} (D \hat{\otimes} U^{\odot}) \xrightarrow{\tau} \mathrm{Hom}(C^{\odot}, D \hat{\otimes} V^{\odot}) \xrightarrow{\mathrm{Hom}(I,\tau)} \mathrm{Hom}(C^{\odot}, \mathrm{Hom}(D^{\odot}, \widehat{V^{\odot}})).$$

Hence, to show the identity, it is enough to evaluate at a in $C^{\odot}$ and b in $D^{\odot}$. Since we have $\rho_{(a \hat{\otimes} b)} = (\rho_1)_a \circ (\rho_2)_b$ and $\phi_{(a \hat{\otimes} b)} = (\phi_2)_b \circ (\phi_1)_a$ with $\phi_i = (\rho_i)^\circ$ and $\phi = (I \hat{\otimes} \phi_2) \circ \phi_1$, it follows from (c) that we have $(\rho^\circ)_{(a \hat{\otimes} b)} = \phi_{(a \hat{\otimes} b)}$. Hence $\phi = \rho^\circ$. Q.E.D.

5.11. THEOREM. *Let C be a topological coalgebra and let V be a topological right C-comodule with structure $\rho: V \to V \hat{\otimes} C$. Then $V^{\odot}$ is a topological left C-comodule with structure $\rho^{\circ}: V^{\odot} \to C \hat{\otimes} V^{\odot}$.*

Proof. We have $(I \hat{\otimes} \Delta) \circ \rho = (\rho \hat{\otimes} I) \circ \rho: V \to V \hat{\otimes} C \to V \hat{\otimes} C \hat{\otimes} C$. Its $^{\circ}$ is by (a) and (d) above

$$(\Delta \hat{\otimes} I) \circ (\rho^{\circ}) = (I \hat{\otimes} (\rho^{\circ})) \circ (\rho^{\circ}): V^{\odot} \to C \hat{\otimes} V^{\odot} \to C \hat{\otimes} C \hat{\otimes} V^{\odot}.$$

Similarly one can prove $(\varepsilon \hat{\otimes} I) \circ (\rho^{\circ}) = I$. Q.E.D.

With the above assumption, for a in $C^{\odot}$, the maps $\rho_a: V \to \hat{V}$ and $(\rho^{\circ})_a: V^{\odot} \to \widehat{V^{\odot}}$ are the actions by a arising from the C-comodules V and $V^{\odot}$. Since $(\rho^{\circ})_a$ is $(\rho_a)^{\odot}$ composed with the canonical map into the completion, and $(\rho_a)^{\odot}$ is the action by a we described above (5.8), it follows that the Ω-topological $C^{\odot}$-module $\widehat{V^{\odot}}$ associated with the C-comodule $V^{\odot}$ is precisely the completion of the Ω-topological $C^{\odot}$-module $V^{\odot}$ of (5.8). Thus we have:

5.12. COROLLARY. *The functor $V \mapsto V^{\odot}$ from right (or left) topological C-comodules into left (or right) topological C-comodules described in* (5.11) *coincides with the functor obtained from* (5.7) *and* (5.8).

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